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ORIGINAL ARTICLE

The impact of the form of the Euler equations for radial flow in cylindrical and spherical coordinates on numerical conservation and accuracy

P. E. Crittenden[1] · **S. Balachandar**[1]

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Abstract The radial one-dimensional Euler equations are often rewritten in what is known as the geometric source form. The differential operator is identical to the Cartesian case, but source terms result. Since the theory and numerical methods for the Cartesian case are well-developed, they are often applied without modification to cylindrical and spherical geometries. However, numerical conservation is lost. In this article, AUSM$^+$-up is applied to a numerically conservative (discrete) form of the Euler equations labeled the geometric form, a nearly conservative variation termed the geometric flux form, and the geometric source form. The resulting numerical methods are compared analytically and numerically through three types of test problems: subsonic, smooth, steady-state solutions, Sedov's similarity solution for point or line-source explosions, and shock tube problems. Numerical conservation is analyzed for all three forms in both spherical and cylindrical coordinates. All three forms result in constant enthalpy for steady flows. The spatial truncation errors have essentially the same order of convergence, but the rate constants are superior for the geometric and geometric flux forms for the steady-state solutions. Only the geometric form produces the correct shock location for Sedov's solution, and a direct connection between the errors in the shock locations and energy conservation is found. The shock tube problems are evaluated with respect to feature location using an approximation with a very fine discretization as the benchmark. Extensions to second order appropriate for cylindrical and spherical coordinates are also presented and analyzed numerically. Conclusions are drawn, and recommendations are made. A derivation of the steady-state solution is given in the Appendix.

Keywords Euler equations · Numerical conservation · Truncation error · Spherical/cylindrical coordinates · Shock · Compressible flow · AUSM$^+$-up

1 Introduction

The Euler equations govern compressible, inviscid, adiabatic flow. Explosive dispersal, inertial confinement fusion, supernovae, and nuclear explosions may be modeled using the Euler equations with radial symmetry. The radial Euler equations can be rewritten in a form which has the identical differential operator as the Cartesian case and the same set of conserved variables. In this form, the geometric information is contained in source terms; hence, it is known as the geometric source form. Because the theory and numerical methods for one-dimensional flow in Cartesian coordinates are well-developed, they are sometimes applied without modification to cylindrical and spherical geometries. However, the geometric source form is not numerically conservative. A scheme is said to be numerically conservative if conserved quantities are maintained globally to machine precision [1]. A discrete, conservative form of the Euler equations can be derived by integrating the governing equations over a control volume. This form will be referred to as the geometric form because the exact volumes and surface areas of the control volumes are retained. A nearly conservative variation of the geometric form may be obtained either by approximating the control volumes to first order or directly from the Euler equations. Since the exact surface areas of the control volumes are retained in the flux terms, it will be called the geometric flux

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✉ P. E. Crittenden
pcritte@ufl.edu

[1] University of Florida, Gainesville, FL, USA

form. In cylindrical coordinates, the geometric and geometric flux forms are identical.

All three forms are identical mathematically as the grid spacing approaches zero, but discretizing the different forms results in different approximations. These different numerical approximations lead to different orders of error in conservation, different truncation rate constants, and as a result lead to different numerical steady-state and time-dependent solutions. It is thought this is primarily due to the geometric source being treated as a source, rather than a flux. Methods are built upon carefully applying conservation laws to finite volumes and the proper treatment of the fluxes. Hence, for the geometric source form, only a portion of the flux is treated properly.

Falle [2] previously derived the numerically conservative form for cylindrical coordinates. Mönchmeyer and Müller [3] derived a form for spherical coordinates, except on a staggered grid of scalar and vector quantities. Li [4] noted that the geometric source form does not satisfy numerical conservation, but did not provide an error estimate. Li also derived the geometric flux form through a change of variables, but could not avoid negative pressures when simulating Sedov's similarity solution for the simulation of point-source explosions [5]. This is probably due to not taking into account a more restrictive initial time step limitation that is necessary to avoid negative internal energies in shock-tube-type problems. Li was primarily interested in developing a higher order method satisfying a weaker conservation property, but also demonstrated a numerically conservative second-order method using an approximate Riemann solver. A numerical comparison was made between forms similar to the geometric and geometric flux forms using Sedov's similarity solution and a different (fifth order) numerical method [6]. They showed a closer agreement between the two forms with respect to shock position than is found here. This is most likely because they model the point source over three cells instead of one. Here, the majority of the error is shown to accrue in the first cell. Toro [7] mentions the geometric flux form used here has some advantages in the context of axisymmetric problems and uses the geometric source form in numerical examples. In [8,9], the geometric flux formulation is used to develop a hybrid finite volume and finite element method in cylindrical and spherical coordinates.

The goal of this paper is to investigate analytically and numerically the impact of the form of the Euler equations with radial symmetry upon numerical conservation and accuracy. In particular, the effect of the form is desired when the advection upwind splitting method with velocity and pressure diffusion (AUSM$^+$-up) [10,11] is used as the flux scheme. As a result of the investigation, some general guidelines are sought on how to best apply AUSM$^+$-up to problems with radial symmetry. Analytically, the order of the numerical conservation error is found for each form, and each form is shown to result in constant total enthalpy for steady flows. Numerically, these properties and the accuracy of each form are evaluated using three types of test problems: steady-state solutions, Sedov's solution, and shock tube problems. Accuracy is evaluated in terms of truncation error, feature location, and agreement with analytical solutions. Both spherical and cylindrical cases are considered.

To extend the spatial results to second order without introducing additional extrema, the monotonic upstream-centered scheme for conservation laws (MUSCL) [12] is used. Cylindrical and spherical geometries also have an impact on this extension. Both [2,3] derived second-order methods and argued that the proper point of evaluation within the cells based upon error analysis and physical grounds is the center of volume, rather than the midpoint of the cell. It is easy to show that if a linear interpolation of the primitive variables is assumed, then the mean value occurs at the center of volume, not the midpoint. In fact, this result does not depend upon the slope of the interpolation; thus, in the limit, as the slope approaches zero, the point of evaluation should also be at the volume center for a first-order method. Since constant properties are assumed for a first-order method, the center of volume would also be the center of mass. Thus, one could also argue for this as the point of evaluation for the momentum equation on physical grounds. Additionally, in [3], the center of volume is shown to be a second-order approximation of the center of mass for any smooth density function. When extending to second-order spatial results, choices can be made as to which primitive variables to interpolate. Liou [10] recommends interpolating density, velocity, and total enthalpy.

To enable testing that total enthalpy is constant for non-trivial steady flows, as well as other properties, a solution representative of all smooth, steady, radial solutions is derived. Clain et al. [13] presented a partially analytic, steady-state, radial solution as part of a broader solution that included swirling flows. Since the solution obtained here is representative of all smooth radial solutions, it is the same as that in [13]. However, their method results in a differential equation that must be solved numerically for pressure, from which the remaining primitive variables are found analytically. Here, the velocity is first found analytically as an implicit function of radius (radius is given explicitly as a function of velocity). Given the velocity, the remaining primitive variables are found analytically.

While conservation does not guarantee accuracy, a non-conservative method can lead to gross inaccuracies including the incorrect positions of shocks, contact interfaces, and other features of the flow [14]. This is shown here for shocks using Sedov's solution, and a direct correlation between numerical conservation and final shock location

is found. Since the exact solution is known, truncation error of the forms is also analyzed using Sedov's solution. This truncation error analysis leads to a different choice in primitive variables to interpolate when the results are extended to second order than is recommended by Liou.

Shock tube problems in spherical and cylindrical geometries result in fairly complicated flows with an outward traveling primary shock, a contact interface, and secondary (and subsequent) shocks. Mass and energy conservation are compared between forms and to the analytical results. The locations of the shocks and contact interfaces are defined to be at points of inflection in the primitive variables [15] and compared between forms over time. An algorithm for determining the feature locations for the unevenly spaced points of evaluation is given. Since there is not an analytical solution, a second-order approximation with a very fine grid is used as a benchmark.

The rest of this paper is organized as follows: The governing equations are presented in the geometric, geometric flux, and geometric source forms in Sect. 2. The AUSM$^+$-up and MUSCL methods are briefly given in Sects. 3 and 4. The error in numerical conservation is found for all of the forms in Sect. 5. In Sect. 6, the enthalpy is shown to be constant for numerically steady flows for all the three forms. In Sect. 7, the numerical conservation errors and constant enthalpy properties are verified using the steady-state solution, and the truncation errors of each form are analyzed numerically. Sedov's similarity solution for the simulation of point and line-source explosions is presented in Sect. 8 and used to further illustrate the numerical conservation properties and truncation errors of the various forms. In particular, the effect of conservation on the shock location is detailed. In Sect. 9, the forms are tested using spherical and cylindrical shock tube problems with respect to conservation and feature location. Results are discussed, and conclusions are drawn in Sect. 10. The derivation of the steady-state solution is given in the Appendix.

2 Forms of the governing equations

The Euler equations for one-dimensional, compressible, inviscid, radial flow may be written as

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^\alpha}\frac{\partial}{\partial r}\left(r^\alpha \rho u\right) = 0, \tag{1}$$

$$\frac{\partial (\rho u)}{\partial t} + \frac{1}{r^\alpha}\frac{\partial}{\partial r}\left(r^\alpha \rho u^2\right) = -\frac{\partial P}{\partial r}, \tag{2}$$

$$\frac{\partial (\rho E)}{\partial t} + \frac{1}{r^\alpha}\frac{\partial}{\partial r}\left(r^\alpha \rho H u\right) = 0, \tag{3}$$

where $\alpha = 0$, 1, or 2, for Cartesian, cylindrical, or spherical coordinates, respectively. In these equations, ρ is the density, u is the one-dimensional velocity, P is the pressure, r is the spatial coordinate, and t is time. For a perfect gas, the total energy, E, and the total enthalpy, H, are

$$E = c_v T + \frac{u^2}{2}, \tag{4}$$

$$H = c_p T + \frac{u^2}{2} = E + \frac{P}{\rho}, \tag{5}$$

where c_v and c_p are the specific heats at constant volume and pressure, respectively, and T is the temperature of the gas.

2.1 Geometric form

Incorporating the pressure into the flux in the momentum equation, the Euler equations can be rewritten in vector form as

$$\frac{\partial \mathcal{U}}{\partial t} + \frac{1}{r^\alpha}\frac{\partial}{\partial r}\left(r^\alpha \mathcal{F}\right) = \frac{\mathcal{N}}{r^\alpha}. \tag{6}$$

The vector of conserved variables is $\mathcal{U} = [\rho,\ \rho u,\ \rho E]^{\mathrm{T}}$, the fluxes are $\mathcal{F} = [\rho u,\ \rho u^2 + P,\ \rho H u]^{\mathrm{T}}$, and the pressure nozzling term is $\mathcal{N} = [0,\ \alpha r^{\alpha-1} P,\ 0]^{\mathrm{T}}$. A numerically conservative scheme may be derived by averaging (6) over a control volume [2]. Initially, a uniform node spacing, $r_{j+1} = r_j + \Delta r$, is assumed with interfaces at $r_{j+1/2} = r_j + \Delta r/2$. Multiplying (6) by $2\alpha\pi r^\alpha \mathrm{d}r$, integrating from $r_{j-1/2}$ to $r_{j+1/2}$, and dividing by the volume of the cell yield the semi-discrete equation

$$\frac{\mathrm{d}\mathbf{U}_j}{\mathrm{d}t} = -\frac{2\alpha\pi}{V_j}\left(r^\alpha_{j+\frac{1}{2}}\mathbf{F}_{j+\frac{1}{2}} - r^\alpha_{j-\frac{1}{2}}\mathbf{F}_{j-\frac{1}{2}}\right) + \frac{2\alpha\pi\,\Delta r}{V_j}\mathbf{N}_j, \tag{7}$$

where

$$\mathbf{U}_j = \frac{2\alpha\pi}{V_j}\int_{r_{j-1/2}}^{r_{j+1/2}} \mathcal{U}(r,t) r^\alpha \mathrm{d}r, \tag{8}$$

is the spatial mean of vector of conserved variables, V_j is the volume of the jth cell, and $\mathbf{F}_{j+1/2}$ is the vector of numerical approximations of the fluxes at the interface. The nozzling term is given by the integral

$$\mathbf{N}_j = \frac{1}{\Delta r}\int_{r_{j-1/2}}^{r_{j+1/2}} \mathcal{N}(r,t)\mathrm{d}r. \tag{9}$$

For numerical calculations (7) will be written as

$$\frac{\mathrm{d}\left(\widehat{r}_j^{\alpha}\mathbf{U}_j\right)}{\mathrm{d}t} = -\frac{r_{j+\frac{1}{2}}^{\alpha}\mathbf{F}_{j+\frac{1}{2}} - r_{j-\frac{1}{2}}^{\alpha}\mathbf{F}_{j-\frac{1}{2}}}{\Delta r} + \mathbf{N}_j, \tag{10}$$

where

$$\widehat{r}_j^{\alpha} \equiv \frac{V_j}{2\alpha\pi\,\Delta r} = \begin{cases} r_j^{\alpha} + \frac{1}{12}\Delta r^2, & \alpha = 2, \\ r_j^{\alpha}, & \alpha = 1. \end{cases} \tag{11}$$

This expression is exact. The second-order term in spherical coordinates, $\Delta r^2/12$, is important when r_j is of the same order as Δr. For the first three cells, if r_j is used instead of $\widehat{r}_j$, there will be errors in the volumes of these cells of 25, 3.57, and 1.32%. Thus, the error decreases rapidly, but is very significant in the first cell. This cell is often the source of difficulty in numerical methods, especially when applied to collapsing flows [3]. If this term is dropped, even for a first-order method, it will be shown that numerical conservation is lost. There is not a similar discarded term in cylindrical coordinates, and no correction is needed. Since the exact values of the volume are used and the surface areas are retained with the fluxes, this form will be called the geometric form.

If a linear variation in the primitive variables is assumed to achieve second-order spatial accuracy, it is easy to show that the mean value occurs at the center of volume regardless of the slope of the interpolation. Therefore, this location will be used in first- and second-order calculations. This also makes sense physically in the momentum equation since for a first-order method, when constant density is assumed, the center of volume is the center of mass. For a first-order method, this only impacts initializing exact solutions and the plotting and interpretation of final results. The center of volume is also a second-order approximation of the center of mass when the density variation is arbitrary [3]. The centers of volume in terms of midpoints of the cell are given by

$$r_{\mathrm{v},j} = \begin{cases} \dfrac{r_j^2 + \frac{1}{12}\Delta r^2}{r_j^2} r_j, & \alpha = 1, \\ \dfrac{r_j^2 + \frac{1}{4}\Delta r^2}{\widehat{r}_j^2} r_j, & \alpha = 2. \end{cases} \tag{12}$$

These are equivalent to the formulas given in [2,3] except that in [3], they are given for node locations in finite difference form and in [2], they are written in terms of Δr. Here, they are written in terms of r_j to emphasize the difference if $r_{\mathrm{v},j}$ is used instead of r_j. As with $\widehat{r}_j$, the differences are the most pronounced for cells near the origin. For the first three cells,

$$r_{\mathrm{v},j} = \begin{cases} \frac{4}{3}r_1, \ \frac{28}{27}r_2, \ \frac{76}{75}r_3, & \alpha = 1, \\ \frac{3}{2}r_1, \ \frac{15}{14}r_2, \ \frac{39}{38}r_3, & \alpha = 2. \end{cases} \tag{13}$$

Unlike $\widehat{r}_j$, this does have an impact in cylindrical coordinates.

Taking the point of evaluation for pressure to be the center of volume and assuming a linear interpolation of the pressure, the pressure nozzling term is given by

$$\mathbf{N}_j = \begin{bmatrix} 0 \\ \alpha \\ 0 \end{bmatrix} \left[r_j^{\alpha-1} P_{\mathrm{v},j} + \left(\widehat{r}_j^{\alpha} - r_{\mathrm{v},j} r_j^{\alpha-1}\right) \left(\frac{\mathrm{d}P}{\mathrm{d}r}\right)_j \right], \tag{14}$$

where $P_{\mathrm{v},j}$ is the pressure at the center of volume of the cell and $\left(\frac{\mathrm{d}P}{\mathrm{d}r}\right)_j$ is the slope of the interpolation. In cylindrical coordinates, this amounts to moving the point of evaluation of the pressure in the pressure nozzling term from the center of volume to the midpoint of the cell using the interpolation. In spherical coordinates, there is also a correction related to the discrete volume since $\widehat{r}_j^2 = r_j^2 + \Delta r^2/12$. Here too, this is the most significant for cells near the origin since as $j \to \infty, \widehat{r}_j \to r_j$ and $r_{\mathrm{v},j} \to r_j$. Thus, since the nozzling term is essentially evaluated at the cell center, using the center of volume as the point of evaluation for the primitive variables, even for the second-order results, will mostly effect the solution only through the initializing exact solutions, interpolation of the primitive variables to achieve second-order, and the interpretation of the results. Given (13), beyond the first few cells, the impact will be minimal. It is worth noting that the pressure nozzling term is similar to nozzling terms in two-phase flow. Specifically, $\mathbf{N}_j$ is the component of the pressure in the radial direction on the lateral faces of the control volume.

2.2 Geometric flux form

Alternately, multiplying (6) by r^{α} gives

$$\frac{\partial\left(r^{\alpha}\mathcal{U}\right)}{\partial t} + \frac{\partial}{\partial r}\left(r^{\alpha}\mathcal{F}\right) = \mathcal{N}, \tag{15}$$

or in semi-discrete form

$$\frac{\mathrm{d}\left(r_j^{\alpha}\mathbf{U}_j\right)}{\mathrm{d}t} = -\frac{1}{\Delta r}\left(r_{j+\frac{1}{2}}^{\alpha}\mathbf{F}_{j+\frac{1}{2}} - r_{j-\frac{1}{2}}^{\alpha}\mathbf{F}_{j-\frac{1}{2}}\right) + \mathbf{N}_j. \tag{16}$$

This form could also be obtained by approximating the volume in the geometric form to first order. Thus, in cylindrical coordinates, it is identical to the geometric form.

Given its similarity to the geometric form, the points of evaluation in each cell are also taken to be the center of volume. This form is numerically conservative in Cartesian and cylindrical coordinates and nearly conservative in spherical coordinates. Since the exact surface areas are retained with the fluxes, it will be called the geometric flux form.

2.3 Geometric source form

The geometric source form comes from separating the r^α factors from the fluxes, using the product rule, and including all of the pressure contribution in the fluxes. Written compactly, it is

$$\frac{\partial \mathcal{U}}{\partial t} + \frac{\partial \mathcal{F}}{\partial r} = \mathcal{S}. \tag{17}$$

Here, the geometric information is contained in the source term, $\mathcal{S} = -\alpha\rho u[1,\ u,\ H]^{\mathrm{T}}/r$. The left-hand side is identical in form to the Cartesian case. Hence, it initially seems that methods developed for the Cartesian case are directly applicable. In semi-discrete form, it is

$$\frac{\mathrm{d}\mathbf{U}_j}{\mathrm{d}t} = -\frac{1}{\Delta r}\left(\mathbf{F}_{j+\frac{1}{2}} - \mathbf{F}_{j-\frac{1}{2}}\right) + \mathbf{S}_j, \tag{18}$$

where $\mathbf{S}_j$ is a numerical approximations of $\mathcal{S}$. If one was to average the geometric source form over a control volume, after one integration by parts of the flux term, the geometric form would result. For this reason, and since it is how the geometric source form is usually presented in the literature, the point of evaluation of primitive variables within each cell is taken as the midpoint.

All three forms may be derived from the Euler equations. The geometric form is obtained by averaging over a control volume and retaining the exact volume of each control volume. The radial factors corresponding to the volumes and surface areas are retained with the conserved variables and fluxes, respectively. For this reason, it is called the geometric form. Alternately, the geometric form may be obtained from the derivation of the Euler equations on a control volume, if the final step, taking the limit as the control volume goes to zero, is not performed. The geometric flux form may be found by multiplying the Euler equations by r^α and then discretizing the result. Since the radial factors corresponding to the surface areas are retained with the fluxes, it is called the geometric flux form. Alternately, the geometric flux form may also be obtained from the geometric form by approximating the control volume to first order. By (11), this only results in a difference in spherical coordinates; thus, the geometric flux and geometric form are identical in cylindrical coordinates. The geometric source form can be obtained by expanding the fluxes using the product rule and placing the terms with geometric information on the right-hand side. Integrating the geometric source form over a control volume also produces the geometric form.

3 AUSM$^+$-up method

The geometric, geometric flux, and geometric source forms of the Euler equations will be compared with regard to various desirable properties. Some observations are general, while others pertain in particular to when the AUSM$^+$-up method, with or without MUSCL, is applied to $\mathbf{F}$. The AUSM methods are based upon a splitting of the genuinely nonlinear and linearly degenerate characteristics. One can show that despite the factor of r^α, the geometric and geometric flux forms have the same characteristics as the Cartesian case. In this section, the AUSM$^+$-up scheme, as given by Liou [11], is briefly presented. The key component of the AUSM$^+$-up is the separation of the flux terms into the convective and pressure flux components. Specifically,

$$\mathbf{F} = \mathbf{F}^{\mathrm{c}} + \mathbf{F}^{p} \equiv \dot{m}\boldsymbol{\Psi}_{\mathrm{L/R}} + \mathbf{P}, \tag{19}$$

where $\dot{m}$ is the mass flux, $\mathbf{P} = [0,\ P,\ 0]^{\mathrm{T}}$ is the pressure flux, and $\boldsymbol{\Psi} = [1, u, H]^{\mathrm{T}}$ is the vector of convected variables. The subscript L/R is chosen based on the sign of the Mach number at the interface. The values to the left of the interface are used if the Mach number at the interface is non-negative and the values to the right are used otherwise. The mass flux is defined by

$$\dot{m}_{j+1/2} = a_{j+1/2} M_{j+1/2} \rho_{\mathrm{L/R}}, \tag{20}$$

where $a_{j+1/2}$ and $M_{j+1/2}$ are the interface speed of sound and interface Mach number, respectively. The subscript L/R on ρ is chosen in the same manner as the subscript on $\boldsymbol{\Psi}$. The interface Mach number is given by

$$M_{j+1/2} = \mathcal{M}^{+}_{(4)}\left(M_{\mathrm{L}}\right) + \mathcal{M}^{-}_{(4)}\left(M_{\mathrm{R}}\right), \tag{21}$$

in which the left and right Mach numbers are defined based on the interface speed of sound:

$$M_{\mathrm{L/R}} = \frac{u_{\mathrm{L/R}}}{a_{j+1/2}}. \tag{22}$$

Here, the interface speed of sound is taken to be

$$a_{j+1/2} = \min\left(\widehat{a}_{\mathrm{L}}, \widehat{a}_{\mathrm{R}}\right), \tag{23}$$

where $\widehat{a}_{\mathrm{L}}$ and $\widehat{a}_{\mathrm{R}}$ are defined by

$$\widehat{a}_{\mathrm{L}} = a^{*2}_{j} / \max\left(a^{*}_{j}, u_{\mathrm{L}}\right), \tag{24}$$

$$\widehat{a}_{\mathrm{R}} = a_j^{*2} / \max\left(a_j^*, -u_{\mathrm{R}}\right). \tag{25}$$

In these equations, a_j^* is the critical speed of sound when the local Mach number is one. For a perfect gas,

$$a_j^{*2} = \frac{2\,(\gamma-1)}{\gamma+1} H_j. \tag{26}$$

This choice for $a_{j+1/2}$ resolves stationary shocks exactly [11].

The functions $\mathcal{M}_{(4)}^{\pm}$ are based on expansions of the non-linear characteristics and are given by

$$\mathcal{M}_{(4)}^{\pm}(M) = \begin{cases} \mathcal{M}_{(1)}^{\pm}, & |M| \geq 1, \\ \mathcal{M}_{(2)}^{\pm}\left[1 \mp 2\mathcal{M}_{(2)}^{\mp}\right], & |M| < 1, \end{cases} \tag{27}$$

where

$$\mathcal{M}_{(1)}^{\pm} = \frac{1}{2}\left(M \pm |M|\right), \quad \mathcal{M}_{(2)}^{\pm} = \pm\frac{1}{4}\left(M \pm 1\right)^2. \tag{28}$$

Similarly, the interface pressure is defined by

$$P_{j+1/2} = \mathcal{P}_{(5)}^{+}\left(M_{\mathrm{L}}\right) P_{\mathrm{L}} + \mathcal{P}_{(5)}^{-}\left(M_{\mathrm{R}}\right) P_{\mathrm{R}}, \tag{29}$$

with

$$\mathcal{P}_{(5)}^{\pm}(M) = \begin{cases} \frac{1}{M}\mathcal{M}_{(1)}^{\pm}, & |M| \geq 1, \\ \mathcal{M}_{(2)}^{\pm}\left(\pm 2 - M \mp 16\widehat{\alpha} M \mathcal{M}_{(2)}^{\mp}\right), & |M| < 1. \end{cases} \tag{30}$$

To this point, if $\widehat{\alpha}$ is taken to be 3/16, the method described is AUSM$^+$. To suppress oscillations near discontinuities, coupling is added between the convective and pressure flux terms, while a scaling is introduced to suppress oscillations in low-Mach number flows. As a result, the equations for the interface mass flux and pressure (20) and (29), become

$$\dot{m} = a_{j+1/2} M_{j+1/2} \rho_{\mathrm{L/R}} + M_P, \tag{31}$$

$$P_{j+1/2} = \mathcal{P}_{(5)}^{+}\left(M_{\mathrm{L}}\right) P_{\mathrm{L}} + \mathcal{P}_{(5)}^{-}\left(M_{\mathrm{R}}\right) P_{\mathrm{R}} + P_u. \tag{32}$$

The scaled pressure difference and convection dissipation terms are

$$M_P = -\frac{\kappa_P}{f_{\mathrm{a}}} \max\left(1 - \overline{M^2}, 0\right) \frac{P_{\mathrm{R}} - P_{\mathrm{L}}}{\rho_{\frac{1}{2}} a_{j+1/2}^2}, \tag{33}$$

$$P_u = -2\kappa_u \mathcal{P}_{(5)}^{+}\left(M_{\mathrm{L}}\right) \mathcal{P}_{(5)}^{-}\left(M_{\mathrm{R}}\right) f_{\mathrm{a}} \rho_{j+1/2} \times a_{j+1/2}\left(u_{\mathrm{L}} - u_{\mathrm{R}}\right), \tag{34}$$

in which $\rho_{j+1/2}$ and $\overline{M^2}$ are the arithmetic means of the density and the square of the Mach numbers to the left and right of the interface. The scaling factor f_{a} is defined by

$$f_{\mathrm{a}} = M_0(2 - M_0), \tag{35}$$

$$M_0 = \min\left[1, \max\left(\overline{M^2}, M_\infty^2\right)\right]. \tag{36}$$

where M_0 is defined in order to remove the scaling in high-speed regions and to avoid divide by zero errors through the use of the limiting Mach number M_∞. Additionally, $\widehat{\alpha}$ in the interfacial pressure fluxes is scaled as

$$\widehat{\alpha} = \frac{3}{16}\left(-4 + 5 f_{\mathrm{a}}^2\right). \tag{37}$$

A method is called pressure balancing if quiescent flows remain steady. In this case $P_{j+1/2} = P_{j-1/2} = P$, where P is the constant pressure. It is trivial to show the geometric source form is pressure balancing. For the geometric and geometric flux forms, since

$$\frac{r_{j+1/2}^{\alpha} - r_{j-1/2}^{\alpha}}{\Delta r} = \alpha r_j^{\alpha-1}, \quad \alpha = 1, 2, \tag{38}$$

where the left- and right-hand sides are the coefficients of the pressure flux and pressure nozzling terms (of the momentum equation), respectively; they too are pressure balancing.

Variations in the AUSM method are denoted by AUSM$^+$ (κ_P, κ_u)-up. From experimentation with test problems, Liou [11] recommends $(\kappa_P, \kappa_u) = (0.25, 0.75)$. If scaling is not used ($f_{\mathrm{a}} = 1$), the scheme is known as the basic AUSM$^+$-up method. The basic version is recommended for shock tube problems where both high- and low-speed flows exist. The method AUSM$^+$-up(0,0) (with $f_{\mathrm{a}} = 1$) simplifies to the AUSM$^+$ method. When the analytical results about properties of the different forms rely on a particular method, those of the simpler AUSM$^+$ method are used. However, the same properties are investigated numerically using AUSM$^+$-up with and without scaling, as appropriate. Alone AUSM$^+$-up is a first-order method. For spatially second-order results, it will be augmented with MUSCL. Unless noted otherwise in the examples, time will be advanced using a third-order, total variation diminishing (TVD), Runge–Kutta method [16]. A method that avoids introducing additional oscillations is known as TVD.

4 MUSCL

The MUSCL [12] is a TVD interpolation method for the interfacial left and right values of the primitive variables. In this work, it will be implemented as outlined in [10] for use with AUSM$^+$-up, except as modified here for cylindrical and spherical coordinates. Additionally, Liou recommends

interpolating (ρ, u, H), while throughout this work, for reasons given in Sect. 8, (ρ, u, P) are interpolated instead.

If a linear variation in a primitive variable W_j is assumed, then the mean value is achieved at the center of volume, $r_{\mathrm{v},j}$, rather than at the midpoint, r_j [2,3]. The slope of the interpolation of W_j to the interfacial values, $W^{\mathrm{L/R}}_{j+\frac{1}{2}}$, is determined by a nonlinear limiting function of the ratio of successive gradients to the left and across the interface for the left facial values and across and to the right of the interface for the right facial values. Specifically,

$$W^{\mathrm{L}}_{j+\frac{1}{2}} = W_j + \left(r_{j+1/2} - r_{\mathrm{v},j}\right) \Phi(g_j) \frac{W_j - W_{j-1}}{r_{\mathrm{v},j} - r_{\mathrm{v},j-1}}, \tag{39}$$

$$W^{\mathrm{R}}_{j+\frac{1}{2}} = W_{j+1} + \left(r_{j+1/2} - r_{\mathrm{v},j+1}\right) \times \Phi\left(\frac{1}{g_{j+1}}\right) \frac{W_{j+2} - W_{j+1}}{r_{\mathrm{v},j+2} - r_{\mathrm{v},j+1}}. \tag{40}$$

The function $\Phi(g_j)$ is the slope limiter and

$$g_j = \left(\frac{W_{j+1} - W_j}{r_{\mathrm{v},j+1} - r_{\mathrm{v},j}}\right) \left(\frac{r_{\mathrm{v},j} - r_{\mathrm{v},j-1}}{W_j - W_{j-1}}\right), \tag{41}$$

is the ratio of successive gradients.

There are various choices for the slope limiter used in the literature. The one used here is called the minmod limiter and is given by

$$\Phi(g) = \max\left[0, \min\left(1, g\right)\right]. \tag{42}$$

The limiter is symmetric and TVD. A symmetric scheme treats forward and backward gradients the same. The net effect of MUSCL combined with the minmod limiter is that the primitive variables are interpolated within each cell using the smallest (in magnitude) of the two gradients calculated using the cell value and either the value in the cell to the left or the cell value to the right. The only exception to this is when the slopes are of opposite sign, in which case the method reduces to first order.

Given two of the interfacial values of the thermodynamic variables, the rest can be found from the closure relations. Once the extrapolated values of the left and right states are known for the primitive variables, the AUSM$^+$-up method presented in the previous section can be applied. For the reasons mentioned before, for the geometric source form, the mean values will be taken to be at the midpoints rather than the centers of volume. Specifically, r_j will be used for the geometric source form instead of $r_{\mathrm{v},j}$, throughout.

5 Numerical conservation

A general conservation law for an invariant quantity in Cartesian, cylindrical, or spherical coordinates can be written as [4]

$$\int_0^{\mathrm{R}} w(t,r) r^\alpha \mathrm{d}r = \int_0^{\mathrm{R}} w(0,r) r^\alpha \mathrm{d}r. \tag{43}$$

Here, w is ρ or ρE and the integral is the total mass, M_{T}, or total energy, E_{T}, in the entire domain. If $\bar{w}^n_j$ is the mean value of w in cell j at time step n, then in discretized form this becomes

$$\sum_{j=1}^{N} \bar{w}^n_j \widehat{r}^\alpha_j \Delta r = \sum_{j=1}^{N} \bar{w}^0_j \widehat{r}^\alpha_j \Delta r. \tag{44}$$

For the momentum equation, the integral of the change in momentum over the domain should equal the external forces. In particular for the Euler equations, it should be the pressure forces on the boundary. Discretely, for radial flow and Euler time stepping, we obtain

$$\sum_{j=1}^{N} \left(\bar{w}^{n+1}_j - \bar{w}^n_j\right) \widehat{r}^\alpha_j \Delta r = F_p \Delta t, \tag{45}$$

where F_p is the pressure multiplied by the perpendicular area. A method is said to be numerically conservative if, accounting for the fluxes at the boundaries (44) and (45) are satisfied to machine precision [1].

5.1 Geometric form conservation

The density and (specific) total energy (denoted as $\bar{w}$) may be written using the discrete geometric form and Euler time stepping as

$$\bar{w}^{n+1}_j = \bar{w}^n_j - \frac{\Delta t}{\Delta r}\left[\left(\frac{r_{j+\frac{1}{2}}}{\widehat{r}_j}\right)^\alpha f_{j+\frac{1}{2}} - \left(\frac{r_{j-\frac{1}{2}}}{\widehat{r}_j}\right)^\alpha f_{j-\frac{1}{2}}\right], \tag{46}$$

where f is the corresponding flux. Starting at the first time step (44) becomes

$$\sum_{j=1}^{N} \bar{w}^1_j \widehat{r}^\alpha_j \Delta r = \sum_{j=1}^{N} \bar{w}^0_j \widehat{r}^\alpha_j \Delta r - \sum_{j=1}^{N} \left(r^\alpha_{j+\frac{1}{2}} f_{j+\frac{1}{2}} - r^\alpha_{j-\frac{1}{2}} f_{j-\frac{1}{2}}\right) \Delta t. \tag{47}$$

However, the second-series telescopes to the boundary values. Thus, the geometric form conserves total mass and energy.

Similarly, for momentum,

$$\sum_{j=1}^{N}\left(\bar{w}_j^1-\bar{w}_j^0\right)\widehat{r}_j^{\alpha}\,\Delta r=-\sum_{j=1}^{N}\left(r^{\alpha}_{j+\frac{1}{2}}f_{j+\frac{1}{2}}-r^{\alpha}_{j-\frac{1}{2}}f_{j-\frac{1}{2}}\right)\Delta t+\sum_{j=1}^{N}n_j\,\Delta r\,\Delta t, \tag{48}$$

where n_j is the momentum component of $\mathbf{N}_j$, and f_j includes the pressure flux. As with mass and energy, the sum of the fluxes (including the pressure flux) telescope.

$$\sum_{j=1}^{N}\left(\bar{w}_j^1-\bar{w}_j^0\right)\widehat{r}_j^{\alpha}\,\Delta r=-\left(r^{\alpha}_{N+\frac{1}{2}}f^{\mathrm{c}}_{N+\frac{1}{2}}-r^{\alpha}_{\frac{1}{2}}f^{\mathrm{c}}_{\frac{1}{2}}\right)\Delta t-\left(r^{\alpha}_{N+\frac{1}{2}}P_{N+\frac{1}{2}}-r^{\alpha}_{\frac{1}{2}}P_{\frac{1}{2}}\right)\Delta t+\sum_{j=1}^{N}n_j\,\Delta r\,\Delta t, \tag{49}$$

where a superscript of c indicates the convective flux. The pressure flux terms are clearly the pressure force at the outer $(N+1/2)$ and inner boundaries $(1/2)$, while n_j is the average over volume j of $\alpha r_j^{\alpha-1}P_j$. Noting that $r^{\alpha}_{j+1/2}-r^{\alpha}_{j-1/2}=\alpha r_j^{\alpha-1}\Delta r$, this is the pressure multiplied by the increase in area in the radial direction. Alternately, it is the component of the pressure force in the radial direction on the lateral sides of the cell. Thus, the geometric form also conserves total momentum.

Note these mass and energy error estimates in no way depend on the method used to calculate the fluxes, the spatial order, the grid size, or the flow. Although the numerical examples use AUSM$^+$-up, it is the form of the equations that results in conservation, not the flux method or its spatial order. The momentum estimate does assume the pressure is exact, so it would depend upon the overall accuracy of the method and could only be checked when the exact solution is known, since the pressure gradient is being converted to momentum. Since the invariance of the total mass and energy is easy to track even without an analytical solution, only their errors are evaluated in the numerical examples.

5.2 Geometric flux form conservation

The geometric flux form only differs from the geometric form in spherical coordinates. In spherical coordinates, the method conserves total mass and energy provided

$$\sum_{j=1}^{N}\left[\left(\frac{r_{j+\frac{1}{2}}}{r_j}\right)^2 f_{j+\frac{1}{2}}-\left(\frac{r_{j-\frac{1}{2}}}{r_j}\right)^2 f_{j-\frac{1}{2}}\right]\widehat{r}_j^{\alpha}=0. \tag{50}$$

Using the definition of $\widehat{r}_j^{\alpha}$,

$$\begin{aligned}\sum_{j=1}^{N}\left[\left(\frac{r_{j+\frac{1}{2}}}{r_j}\right)^2 f_{j+\frac{1}{2}}-\left(\frac{r_{j-\frac{1}{2}}}{r_j}\right)^2 f_{j-\frac{1}{2}}\right]\widehat{r}_j^{\alpha}&=\frac{\Delta r^2}{12}\sum_{j=1}^{N}\left[r^2_{j+\frac{1}{2}}\left(\frac{r^2_{j+1}-r^2_j}{r^2_j r^2_{j+1}}\right)f_{j+\frac{1}{2}}\right],\\&=\mathcal{O}\left[\Delta r^2\ln(\Delta r)\right].\end{aligned} \tag{51}$$

The above calculation assumes the origin is contained in the domain. If the domain is bounded away from the origin, one may show similarly that the conservation error is at most $\mathcal{O}\left(\Delta r^2\right)$ in each time step. This makes sense since most of the discrepancy between the geometric flux form and the geometric form is near the origin. No assumption about the flux scheme or its spatial order has been made in (51). Thus, the conservation order is independent of the flux scheme used. The conservation error rate constant, because of the presence of $f_{j+\frac{1}{2}}$ in the estimate, may depend on the flux method.

For the momentum equation, since the pressure fluxes are treated identically to the convective fluxes, they have an error of order $\mathcal{O}\left[\Delta r^2\ln(\Delta r)\right]$. Likewise, the sum of the nozzling terms contain an error of order $\mathcal{O}\left[\Delta r^2\ln(\Delta r)\right]$.

5.3 Geometric source form conservation

For the geometric source form of the equations, the conserved variables at the next time step may be written in discrete form as

$$\bar{w}_j^{n+1}=\bar{w}_j^n-\frac{\Delta t}{\Delta r}\left(f_{j+\frac{1}{2}}-f_{j-\frac{1}{2}}\right)-\alpha\Delta t\frac{f_j}{r_j}. \tag{52}$$

It should be noted that $f_j=[\rho_j u_j,\rho u_j^2,\rho_j u_j H_j]$ are the cell-centered values of the quantities used in approximating the fluxes, $f_{j\pm\frac{1}{2}}$, at the cell boundaries.

Again starting at the first time step,

$$\sum_{j=1}^{N}\bar{w}_j^1\widehat{r}_j^{\alpha}\,\Delta r=\sum_{j=1}^{N}\bar{w}_j^0\widehat{r}_j^{\alpha}\,\Delta r-\sum_{j=1}^{N}\left[\frac{\Delta t}{\Delta r}\left(f_{j+\frac{1}{2}}-f_{j-\frac{1}{2}}+\Delta r\alpha\frac{f_j}{r_j}\right)\right]\widehat{r}_j^{\alpha}\,\Delta r. \tag{53}$$

For cylindrical coordinates, using the definition of $\widehat{r}_j$ and the boundary conditions,

$$\sum_{j=1}^{N}\left(f_{j+\frac{1}{2}}-f_{j-\frac{1}{2}}+\Delta r\alpha\frac{f_j}{r_j}\right)\widehat{r}_j^{\alpha}=-\Delta r\sum_{j=1}^{N}\left(f_{j+\frac{1}{2}}-f_j\right). \tag{54}$$

Since $f_{j+\frac{1}{2}}$ is an approximation of f_j at $r_j+\Delta r/2$, the error in numerical conservation will normally be $\mathcal{O}(\Delta r)$ regardless of the order of the method or the flux scheme.

In the spherical case,

$$\begin{aligned}&\sum_{j=1}^{N}\left(f_{j+\frac{1}{2}}-f_{j-\frac{1}{2}}+\Delta r\alpha\frac{f_j}{r_j}\right)\widehat{r}_j^{\alpha}\\&=\sum_{j=1}^{N}\left[r_j^2\left(f_{j+\frac{1}{2}}-f_{j-\frac{1}{2}}\right)+2r_j\Delta r f_j\right]\\&\quad+\frac{\Delta r^2}{12}\sum_{j=1}^{N}\left(f_{j+\frac{1}{2}}-f_{j-\frac{1}{2}}+2\Delta r\frac{f_j}{r_j}\right),\end{aligned} \tag{55}$$

$$\begin{aligned}&=-2\Delta r\sum_{j=1}^{N}r_j\left(f_{j+\frac{1}{2}}-f_j\right)\\&\quad-\Delta r^2\sum_{j=1}^{N}\left[f_{j+\frac{1}{2}}-\frac{1}{6}\frac{f_j}{j-\frac{1}{2}}\right].\end{aligned} \tag{56}$$

Assuming $f_{j+\frac{1}{2}}-f_j=\mathcal{O}(\Delta r)$, again the error bound is $\mathcal{O}(\Delta r)$. The order of the numerical conservation error for the geometric source form is significantly inferior to the geometric and geometric flux forms. As with the geometric flux form, the dependence of the error on the flux scheme or its spatial order is limited to the conservation error rate constant.

For the momentum equation, the discrete approximation of the integral pressure fluxes is

$$\begin{aligned}-\sum_{j=1}^{N}\left(P_{j+\frac{1}{2}}-P_{j-\frac{1}{2}}\right)\widehat{r}_j^{\alpha}&=-\hat{r}_N^{\alpha}P_{N+1/2}+r_1^{\alpha}P_{1/2}\\&\quad-\sum_{j=1}^{N-1}\left(r_j^{\alpha}-r_{j+1}^{\alpha}\right)P_{j+1/2}.\end{aligned} \tag{57}$$

For the cylindrical case, this may be rewritten as

$$\begin{aligned}-\sum_{j=1}^{N}\left(P_{j+\frac{1}{2}}-P_{j-\frac{1}{2}}\right)\widehat{r}_j&=r_{1/2}P_{1/2}-r_{N+1/2}P_{N+1/2}\\&\quad+\frac{\Delta r}{2}\left(P_{1/2}+\sum_{j=1}^{N-1}2P_{j+1/2}+P_{N+1/2}\right),\end{aligned} \tag{58}$$

which can be viewed as the forces at the inner and outer radius plus an approximation of the forces on the lateral boundaries via the trapezoid rule. The accuracy of the trapezoid rule is $\mathcal{O}(\Delta r^3)$, so the conservation error would be the $\mathcal{O}(\Delta r)$ as previously identified for the convective fluxes.

Similarly, for the spherical coordinates,

$$\begin{aligned}-\sum_{j=1}^{N}\left(P_{j+\frac{1}{2}}-P_{j-\frac{1}{2}}\right)\widehat{r}_j^2&=\widehat{r}_{1/2}^2P_{1/2}-\widehat{r}_{N+1/2}P_{N+1/2}\\&\quad+\Delta r\left(r_{1/2}P_{1/2}+\sum_{j=1}^{N-1}2r_{j+1/2}P_{j+1/2}\right.\\&\quad\left.+r_{N+1/2}P_{N+1/2}\right),\\&\quad+\frac{\Delta r^2}{3}\left(P_{1/2}-P_{N+1/2}\right),\end{aligned} \tag{59}$$

which again is the forces at the inner and outer radius and an approximation of the forces on the lateral boundaries, plus an additional error term. Again, the conservation error is $\mathcal{O}(\Delta r)$.

6 Constant enthalpy in steady flows

For steady flows, the total enthalpy remains constant. The proof is as follows. Provided the flow is steady, assuming that $u_j>0$ for all j, the geometric or geometric flux form's continuity equation gives

$$\left(r+\frac{\Delta r}{2}\right)^{\alpha}a_{j+\frac{1}{2}}M_{j+\frac{1}{2}}\rho_j=\left(r-\frac{\Delta r}{2}\right)^{\alpha}a_{j-\frac{1}{2}}M_{j-\frac{1}{2}}\rho_{j-1}. \tag{60}$$

Similarly, for the steady case, the energy equation yields

$$\begin{aligned}&\left(r+\frac{\Delta r}{2}\right)^{\alpha}a_{j+\frac{1}{2}}M_{j+\frac{1}{2}}\rho_jH_j\\&=\left(r-\frac{\Delta r}{2}\right)^{\alpha}a_{j-\frac{1}{2}}M_{j-\frac{1}{2}}\rho_{j-1}H_{j-1}.\end{aligned} \tag{61}$$

However, since the quantities multiplying the total enthalpies were just shown to be equal, $H_j=H_{j+1}$ for $j=0,1,2,\ldots,N$. Note that the factors $\left(r\pm\frac{\Delta r}{2}\right)^{\alpha}$ are proportional to the surface areas of the cells in the corresponding coordinate systems, so the conservation equation is really just giving that in the steady case that the mass flowing in equals the mass flowing out. As a result, regardless of the assumptions about the direction of u_j, similar, equal terms will multiply H_j and H_{j-1} and the same result will follow. The geometric source form of the Euler equations applied to steady flows also preserves constant enthalpy. The proof is

similar. Liou [11] proved this same property for the Cartesian case.

7 Steady-state solution

In Cartesian coordinates, steady, one-dimensional flows are simply constant property flows. For radial flows, a non-trivial solution can be derived (see Appendix) by assuming the solution exists at a single point. In cylindrical or spherical coordinates, one may show if $\rho(r_\mathrm{R}) = \rho_\mathrm{R}$, $u(r_\mathrm{R}) = u_\mathrm{R}$, and $P(r_\mathrm{R}) = P_\mathrm{R}$, that for steady flows

$$\frac{u}{u_\mathrm{R}}\left[\frac{2c_2 - \left(\frac{u}{u_\mathrm{R}}\right)^2}{2c_2 - 1}\right]^{\frac{1}{\gamma-1}} = \left(\frac{r}{r_\mathrm{R}}\right)^{-\alpha}, \tag{62}$$

where

$$c_2 = \frac{\gamma}{\gamma - 1}\mathrm{Eu} + \frac{1}{2}. \tag{63}$$

In this expression, $\mathrm{Eu} = P_\mathrm{R}/(\rho_\mathrm{R} u_\mathrm{R}^2)$ is the Euler number. Given u_R, the density may be found from conservation of mass and the pressure from the constant enthalpy condition. The temperature may then be found from the ideal gas law. The only assumption used in deriving the solution is that the density, pressure, and velocity exist at some reference radius. Therefore, this solution is representative of all smooth, radial, steady-state solutions.

7.1 Spherical steady-state results

The density, pressure, velocity, and temperature for the steady-state solution in spherical coordinates consistent with air treated as an ideal gas ($\gamma = 1.4$) and $\mathrm{Eu} = 858.13$ are plotted in Fig. 1 for $\widetilde{r} \in [0.24, 1]$. In the figure, the variables are scaled by their reference values except $\widetilde{u} = u/a$, where a is the speed of sound at the reference temperature. That is $\widetilde{r} = r/r_\mathrm{R}$, $\widetilde{\rho} = \rho/\rho_\mathrm{R}$, $\widetilde{P} = P/P_\mathrm{R}$, and $\widetilde{T} = T/T_\mathrm{R}$. The time scale is taken to be $t_\mathrm{R} = r_\mathrm{R}/u_\mathrm{R} = 0.1$.

None of the forms preserve the exact steady-state solution. Instead, they evolve over time to different, but nearby numerically steady solutions. The differing solutions arise because, although all forms were shown to have solutions with constant enthalpy and all the forms satisfy the ideal gas equation, the continuity and momentum equations are discretized differently. Since the boundary conditions are set to that of the exact solution, they set the eventual constant enthalpy to the exact value for all three forms. The geometric flux form was shown to satisfy conservation of mass to a higher order than the geometric source form, while the geometric form is exact with respect to conservation.

Both the evolution of the solutions and the final numerical steady solutions may be used to analyze accuracy of the different forms. For this example, the geometric flux form is not analyzed separately, except with respect to conservation, because with the innermost radius of $\widetilde{r} = 0.24$, there is very little difference between it and the geometric form.

7.1.1 Local truncation error

The local truncation error is defined as the error in a single step of the numerical method applied to the exact solution. Since the exact solution is known and set as the initial condition, this error can be found numerically. The boundary (ghost nodes) and initial conditions are set to the analytical solution for $\widetilde{r}_i < \widetilde{r} < \widetilde{r}_o$. Using Euler time-advancing, the truncation error for a primitive variable W at time step n and cell j can be viewed as

$$\begin{aligned}\mathcal{T}\left(W_j^n\right) &= \frac{\partial W}{\partial t} + \mathcal{L}(W) \\ &\quad - \left[\frac{W_j^{n+1} - W_j^n}{\Delta t} + \mathcal{A}_{\mathrm{G/GS}}\left(W_j^n\right)\right], \\ &= K_t \Delta t^\ell + K_r \Delta r^k,\end{aligned} \tag{64}$$

where $\mathcal{L}$ is the original spatial operator from the Euler equations and $\mathcal{A}_{\mathrm{G/GS}}$ is its numerical approximation under the AUSM$^+$-up method using either the geometric (G) or geometric source form (GS). The exponent ℓ is the temporal order of convergence, and k is the spatial order of convergence, while K_t and K_r are the rate constants. Since at the first time step of the numerical method, W_j^0 is the exact solution and $\mathcal{L}(W) = 0$ since it is steady, we have

$$\mathcal{A}_{\mathrm{G/GS}}\left(W_j^1\right) = K_r \Delta r^k. \tag{65}$$

Letting $W_j(r|\Delta r)$ and $W_j(r|\Delta r/2)$ be the numerical approximations of a primitive variable after one time step using two different resolutions and $W_j(r)$ be the analytical solution,

$$k = \frac{\ln\left|\frac{W_j(r|\Delta r) - W_j(r)}{W_j(r|\Delta r/2) - W_j(r)}\right|}{\ln(2)}. \tag{66}$$

Given the input data above, AUSM$^+$(0,0) and AUSM$^+$-up(0.25,0.75) ($M_\infty = 0.7$) were run using both the geometric and geometric source formulations for one time step of 5×10^{-6} with $\Delta\widetilde{r} = 0.01$, and 0.005. The order of convergence for almost all of the primitive variables is near one. The only exception is the density when using the geometric source form without numerical diffusion, which at least, initially, is diverging near $\widetilde{r} = 1$ (see Fig. 2). The grid was shifted $\Delta\widetilde{r}/2$ to the left for the finer discretization to maintain the same

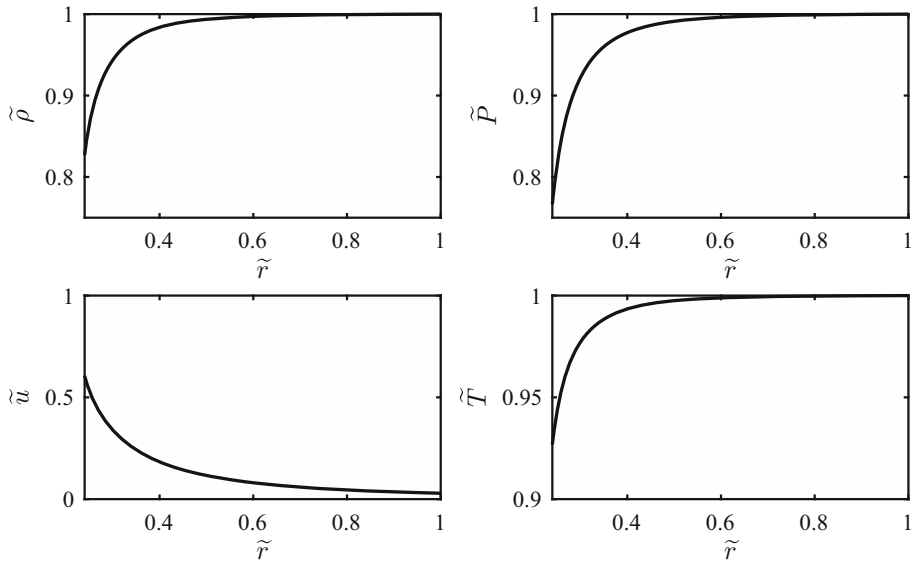


Fig. 1 Analytical steady-state solution in spherical coordinates given in (62) and (63) ($\alpha = 2$)

cell interfaces. Numerical diffusion made the convergence order more uniformly one for density and the other primitive variables. Although the convergence orders are the worst near $\widetilde{r} = 1$, the error is the smallest in that region for all of the primitive variables. As an example, the relative error in the density after one time step is plotted in Fig. 3. It should be noted that while the convergence order is independent of the time step, the relative error is directly proportional to it (when Euler time stepping is used). Numerical diffusion, although it makes the convergence order more uniform and reduces oscillations, increases the truncation errors in the primitive variables. The maximum error in density (near $\widetilde{r} = 0.24$) more than doubles for the geometric source form and somewhat less than doubles for the geometric form. As $\widetilde{r} \to 1$, the errors become quite small, but those using numerical diffusion are orders of magnitude larger than those without.

The ratios of the rate constants with and without numerical diffusion are plotted in Fig. 4. The spikes in some of the curves near $\widetilde{r} = 0.24$ are due to the different forms passing through the exact solution at slightly different locations. Without numerical diffusion, the ratio of the rate constants for density is about 1.5, while for the rest of the primitive variables, the ratios are closer to three across most of the domain.

Using numerical diffusion increases the advantage with respect to the rate constant of the geometric form for density and reduces the advantage for pressure and temperature from roughly three to about two (see Fig. 4). Since pressure becomes nearly constant as $\widetilde{r} \to 1$, the rate constant ratio for u is virtually unaffected by numerical diffusion for $\widetilde{r} > 0.6$ [see (33)]. With or without numerical diffusion, the advantages with respect to rate constants of the geometric form over the geometric source form are independent of $\Delta\widetilde{r}$.

When AUSM$^+$ is augmented with MUSCL, the convergence order of both forms is almost uniformly two, with or without numerical diffusion (see Fig. 2). The relative error in density is reduced by about a factor of 100 (see Fig. 5). Increasing to second order reduces the rate constant advantages of the geometric form (see Fig. 6). Again, the rate constant ratios are independent of the grid spacing, and for the second-order results, they are less sensitive to numerical diffusion. Numerical diffusion increases the rate constant ratios for all but velocity, which again is unaffected for roughly $\widetilde{r} > 0.6$. Evaluating at the volume center versus the midpoint has a negligible effect. For example, the peak values in the rate constant ratios are changed by less than 0.1% if the midpoints of the cells are used as the points of evaluation.

7.1.2 Conservation of mass and energy

Over time, both forms converged to different numerical steady-state solutions, relatively close to the analytical solution. The total mass, M_T, and total energy, E_T, in the domain were calculated as per (44) and tracked as a function of time from $\widetilde{t} = 0$ to $\widetilde{t} = 1$. The steady-state solution should have constant mass flux throughout the domain. In particular, as much mass should be flowing in from the left as is flowing out to the right. This was checked for all three forms and found to not be the case. That is $\widetilde{r}^\alpha_{N+1/2}\widetilde{\rho}_N u_{N+1/2} \neq \widetilde{r}^\alpha_{1/2}\widetilde{\rho}_0 u_{1/2}$, where $\widetilde{\rho}_0$ is the value at the ghost node to the left and $\widetilde{r}_{1/2} = 0.24$. Numerical conservation is a property of the interior points. Therefore, the total mass was adjusted to compensate for this error in the mass flux at the boundaries. There does not seem to be an equitable method to correct the geometric source form for errors caused by the boundary conditions because the geometric source term is due to both

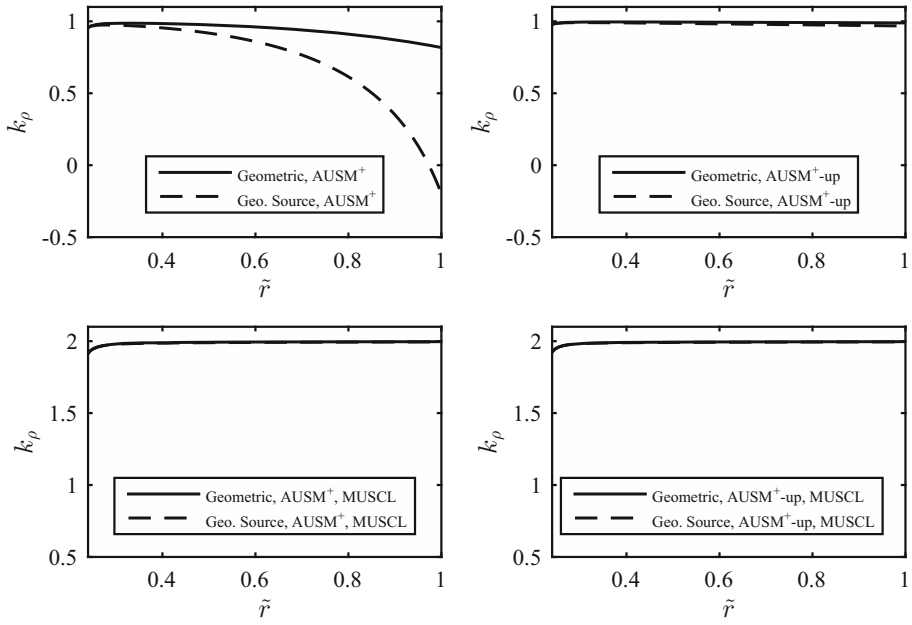


Fig. 2 The convergence order of density, k_ρ, to the spherical steady-state solution using the geometric and geometric source forms. Plots without numerical diffusion are labeled AUSM$^+$. Plots with numerical diffusion are labeled AUSM$^+$-up. Numerical diffusion makes the first-order results more uniformly one. MUSCL is used to achieve second-order

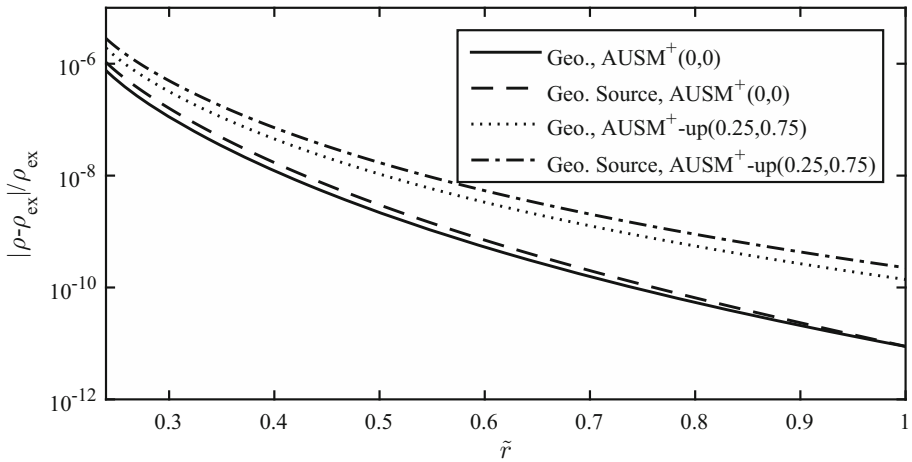


Fig. 3 Absolute relative error in density at initial time step of spherical steady-state solution using the geometric and geometric source forms

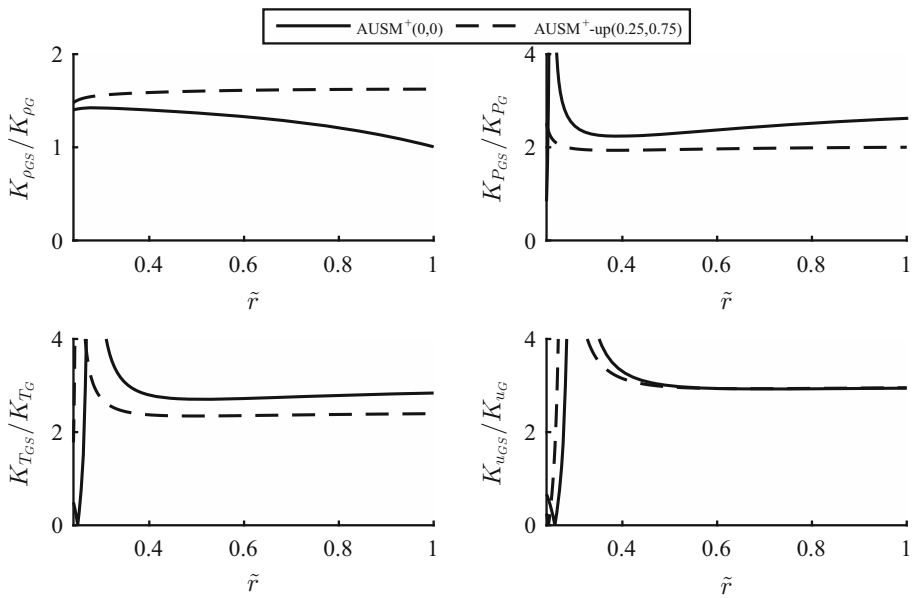


Fig. 4 Initial ratios of rate constants (geometric source:geometric) for spherical steady-state solution using AUSM$^+$(0,0) and AUSM$^+$-up(0.25,0.75)

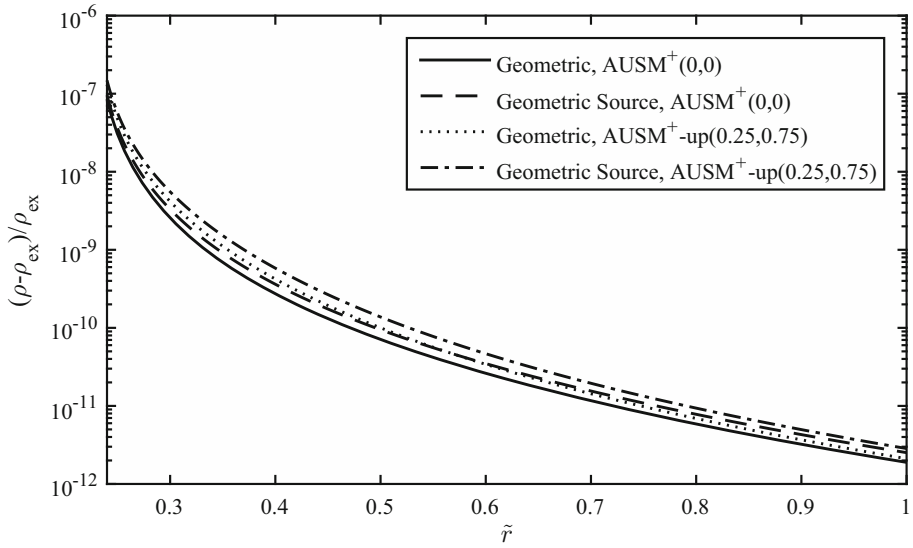


Fig. 5 Absolute relative error in density at initial time step of spherical steady-state solution using the geometric and geometric source forms with MUSCL

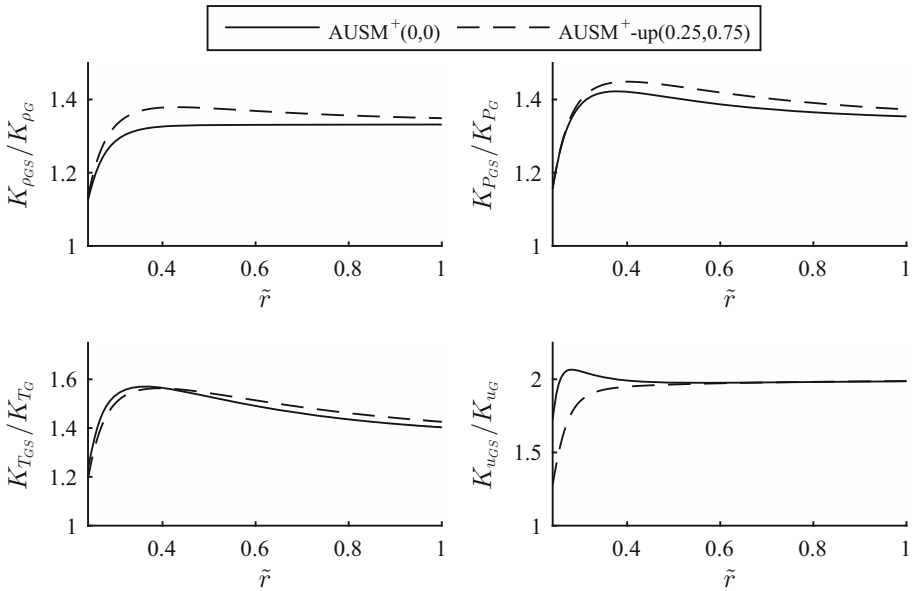


Fig. 6 Initial ratios of rate constants (geometric source:geometric) for spherical steady-state solution using AUSM$^+$(0,0) and AUSM$^+$-up(0.25,0.75) with MUSCL

flux into the left of the cell and out of the right of the cell. As a result, the geometric source errors were only adjusted by the Cartesian fluxes out of the boundaries without considering the geometric source terms.

In Fig. 7, the error in the conservation of mass is plotted as a function of $\Delta\widetilde{r}$ for all of the forms, both with and without numerical diffusion. The relative errors in the total mass are plotted after steady state was reached ($\widetilde{t} = 1$) for three different spatial step sizes, $\Delta\widetilde{r} = 0.0005,\ 0.001,\ 0.002$. Plots at earlier times are quite similar. As can be seen in Fig. 7, the geometric form is clearly superior. The geometric flux form is not quite as good by this measure without MUSCL, but is still several orders of magnitude better than the geometric source form. Using MUSCL, the second-order results for the geometric flux form reach round-off error for the grid sizes used, and as a result, the errors are comparable to the geometric form.

The numerical orders of convergence for the total mass using the two finest discretizations are given in Table 1 and closely agree with the theoretical results. In the table, the values given were obtained using the Runge–Kutta TVD method [16]. If Euler time stepping is used for the geometric flux form with AUSM$^+$(0,0), then the order is 2.99 compared to the theoretical value of 2.00 from Sect. 5.2. The order is higher in this particular example because it is a steady-state solution. Specifically, since it is steady state, the net flux out of each cell divided by Δr, $(r^2_{j+\frac{1}{2}} f_{j+\frac{1}{2}} - r^2_{j-\frac{1}{2}} f_{j-\frac{1}{2}})/\Delta r$, is zero to first order. Thus (50) becomes $\mathcal{O}(\Delta r^3)$. The Runge–Kutta time-stepping technique improves the order of convergence even further. When using MUSCL, the geometric form and

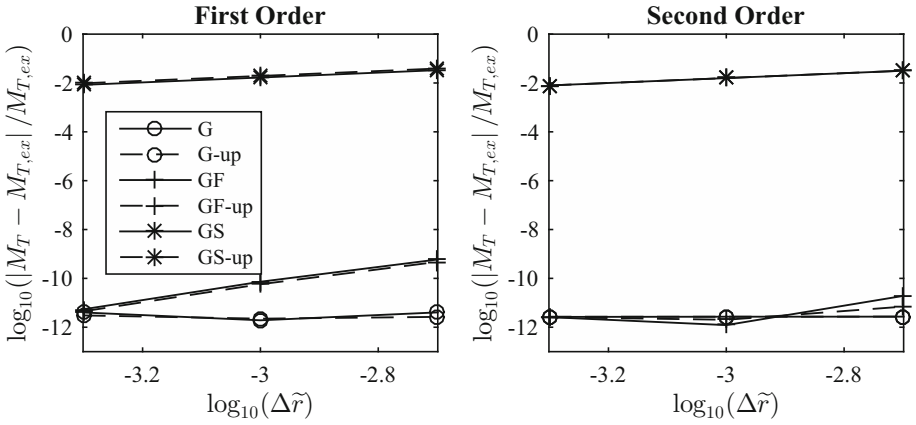


Fig. 7 Error in total mass at steady state in spherical coordinates using the geometric form (G), geometric flux form (GF) and the geometric source form (GS) with the basic AUSM$^+$(0,0). The notation (-up) indicates numerical diffusion, AUSM$^+$-up(0.25,0.75), $M_\infty = 0.7$. Second-order results are via MUSCL

geometric flux form errors in total mass are at machine precision. From Sect. 5.1, this was expected for the geometric form, while for the geometric flux form, it is thought to be due to a combination of the effects of the steady-state solution, the fine grid, and the second-order spatial method. For a second-order spatial method at steady state, theoretically, the error should be $\mathcal{O}(\Delta r^4)$. The orders reported in the table for the geometric source form agree with the results of Sect. 5.3, both with respect to conservation error order and in that they are independent of the spatial order of the method.

If the total mass is not adjusted for the net (erroneous) flux out of the boundaries, then all three forms have comparable error levels with respect to conservation. Hence, ghost nodes may not be the best method of enforcing the boundary conditions. Liou [10] has given alternate characteristic-based methods for reflection boundaries, but deriving such an extension for the steady case is beyond the scope of this current paper. This is not an issue for the remaining examples, which have zero convective flux at the boundaries.

7.1.3 Constant total enthalpy

It was shown in Sect. 6 that the numerical solutions have constant total enthalpy for all three forms. Furthermore, this constant is set to the same constant value as the analytical solution by the values at the ghost nodes. However, due to the different discretizations of the continuity and momentum equations, this is achieved through different combinations of ρ, u, and P. The mismatch between the fluxes of the analytical and approximate solutions causes "waves" of enthalpy to move through the domains resulting in local errors in the enthalpy as they pass. Plots of the relative errors in total enthalpy for a few early times are given in Fig. 8 for the geometric and geometric source forms. The "wave" for geometric form has a smaller amplitude, but by the time steady state is reached, the waves exit to the right of the domain and the errors in the final total enthalpy are near round-off error (through different combinations of ρ, u, and P).

7.1.4 Root-mean-square error of steady-state solutions

The solutions were run for $\Delta\widetilde{r} = 0.005$ until $\widetilde{t} = 1$, which was well after steady state was reached. The second-order results without numerical diffusion are highly oscillatory especially near the boundaries. Although the error is less than the first-order results, it is $\mathcal{O}(10^{-5})$, making it well above round-off error and unacceptable. Even the first-order solutions without numerical diffusion have oscillations near $\widetilde{r} = 1$. Given the locations of the oscillations, they are probably due to the implementation of the boundary conditions through ghost nodes. However, they are eliminated by numerical diffusion. While numerical diffusion eliminates the oscillations, it increases the maximum error. The root-mean-square (rms) errors of the numerical steady-state solutions when using the geometric source and geometric forms and their ratios are summarized in Table 2. By this measure when using AUSM$^+$(0,0), the geometric form is over twice as accurate with respect to density and pressure and over ten times as accurate for velocity and temperature. If instead numerical diffusion is used, the advantage of the geometric form to the geometric source form increases to four times as much for density and pressure, while it is diminished slightly for the velocity and temperature. When using the geometric form, the first-order rms errors for density and pressure and all of the rms errors for the second-order results decrease when numerical diffusion is used, probably due to the elimination of oscillations. The error in the velocity and temperature using AUSM$^+$ for the geometric form and all of the rms errors for the geometric source form are increased by numerical diffusion and in most cases nearly doubled.

Table 1 Numerical spatial convergence orders of total mass for all three forms using AUSM$^+$(0,0) and AUSM$^+$-up(0.25,0.75), $M_\infty = 0.7$

Form	AUSM$^+$	AUSM$^+$-up	AUSM$^+$ & MUSCL	AUSM$^+$-up and MUSCL
Geometric	Exact	Exact	Exact	Exact
Geometric flux	3.75	3.69	Exact	Exact
Geometric source	1.00	1.00	1.00	1.00

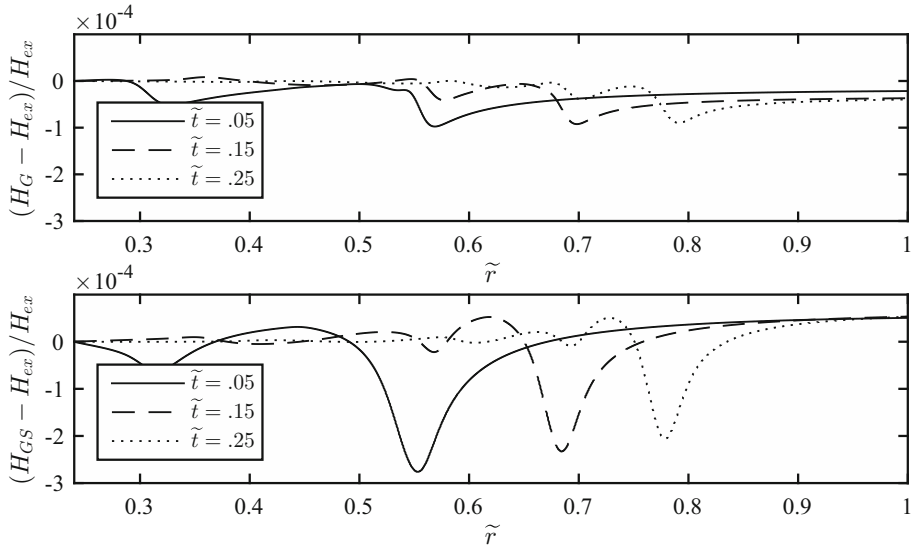


Fig. 8 Relative error in total enthalpy in spherical coordinates as steady state is approached using AUSM$^+$(0,0) and $\Delta\widetilde{r} = 0.005$

Table 2 Root-mean-square errors of primitive steady-state variables and error ratios in spherical coordinates

	Density	Pressure	Velocity	Temperature
AUSM$^+$(0,0)				
Geometric	2.10e−04	2.06e−04	3.71e−05	5.84e−06
Geometric source	4.80e−04	5.24e−04	4.13e−04	7.77e−05
Ratio (GS:G)	2.29	2.54	11.15	13.30
AUSM$^+$-up(0.25,0.75)				
Geometric	1.99e−04	1.98e−04	7.63e−05	1.21e−05
Geometric source	8.02e−04	8.78e−04	6.85e−04	1.33e−04
Ratio (GS:G)	4.03	4.43	8.98	11.03
AUSM$^+$(0,0) & MUSCL				
Geometric	3.48e−06	4.31e−06	2.81e−06	1.28e−06
Geometric source	3.96e−06	4.85e−06	3.67e−06	1.38e−06
Ratio (GS:G)	1.14	1.13	1.31	1.08
AUSM$^+$(0.25,0.75) & MUSCL				
Geometric	2.53e−06	2.82e−06	2.39e−06	6.00e−07
Geometric source	4.42e−06	4.97e−06	4.20e−06	9.54e−07
Ratio (GS:G)	1.75	1.76	1.76	1.59

7.2 Cylindrical steady-state results

A cylindrical case with was run on the same domain except with Eu = 53.633. The Euler number was decreased to obtain roughly the same amount of increase in velocity as in the spherical case. The results are qualitatively similar, but the advantage of the geometric form is not as pronounced.

The orders of convergence are more uniform, even without numerical diffusion. The geometric form still has a distinct, although somewhat reduced advantage with respect to rate constants. Overall, the rate constant advantages are around 20% less than those for the spherical case. This is not entirely unexpected since there are in general three sources of error for the geometric source form: the approximation of the discrete

Table 3 Root-mean-square errors of primitive steady-state variables and error ratios in cylindrical coordinates

	Density	Pressure	Velocity	Temperature
AUSM$^+$(0,0)				
Geometric	1.11e−04	1.18e−04	6.86e−05	1.24e−05
Geometric source	2.30e−04	2.52e−04	2.05e−04	3.72e−05
Ratio (GS:G)	2.08	2.14	2.99	3.00
AUSM$^+$-up(0.25,0.75)				
Geometric	2.28e−04	2.50e−04	1.88e−04	3.56e−05
Geometric source	4.42e−04	4.91e−04	4.10e−04	7.61e−05
Ratio (GS:G)	1.94	1.97	2.18	2.14
AUSM$^+$(0,0) & MUSCL				
Geometric	1.01e−06	1.25e−06	8.77e−07	3.61e−07
Geometric source	1.11e−06	1.36e−06	1.05e−06	3.78e−07
Ratio (GS:G)	1.10	1.08	1.20	1.05
AUSM$^+$(0.25,0.75) & MUSCL				
Geometric	1.15e−06	1.29e−06	1.03e−06	2.19e−07
Geometric source	1.55e−06	1.74e−06	1.43e−06	3.03e−07
Ratio (GS:G)	1.34	1.35	1.40	1.39

volume, the treatment of part of the fluxes as source terms, and the use of the cell centers instead of the centers of volume. In cylindrical coordinates, there is no approximation of the cell volume and given that the inner radius is 0.24, there is little difference between the centers of volume and the cell centers. Thus, most of the differences in errors in cylindrical coordinates for this test case can be attributed to the treatment of the fluxes.

As with the spherical case, the errors in steady-state pressure and velocity are highly oscillatory when AUSM$^+$ is coupled with MUSCL without numerical diffusion. The rms errors in the primitive variables and their ratios are summarized in Table 3. The rms values are roughly of the same order as those for the spherical case. In general, the ratios show an advantage for the geometric form. However, comparing Tables 2 and 3, the advantage of the geometric form is smaller in cylindrical coordinates than it was for spherical coordinates. Numerical diffusion, while it reduces the oscillations in the results, roughly doubles the rms errors for the first-order results, while for the second-order results, it is increased for all but velocity. Numerical diffusion decreases the rate constant advantage of the geometric form for the first-order results and increases it for the second-order results. All of the ratios are less than they were for the spherical case.

8 Sedov solution

Sedov derived an analytical solution for point or line sources of energy [5]. As with the steady-state solution, this provides another opportunity to check the truncation error and conservation properties of the three forms. Unlike the steady-state solution, there is a strong shock. Predicting the correct shock location provides another measure for comparing the forms of the Euler equations. First Sedov's solution is briefly presented. The presentation is largely from [17] and [18], where they detail a family of solutions depending on an exponentially varying background density and include special cases resulting from singularities. Here, the presentation is condensed to only cover a constant background density and the "standard" (non-singular) case. A modification of their algorithm for producing plots of the primitive variables based upon the solution is also given.

8.1 Similarity solution

Assuming a point charge of energy, $\mathcal{E}$, is released at the origin in a background density of ρ_1, a strong shock results which produces a large pressure discontinuity, $P_2/P_1 \gg (\gamma+1)/(\gamma-1)$. Subscripts of 1 will be used to indicate the undisturbed values of the primitive variables, while subscripts of 2 indicate the values just behind the shock. Neglecting P_1, a similarity solution can be found to the Euler equations [19].

The position of the strong shock is

$$r_2 = \left(\frac{\mathcal{E}t^2}{\rho_1 \delta}\right)^{1/(\alpha+3)}, \tag{67}$$

with velocity

$$U \equiv \frac{\mathrm{d}r_2}{\mathrm{d}t} = \frac{2}{\alpha+3}\frac{r_2}{t}. \tag{68}$$

The non-dimensional energy, δ, is given by the integral

$$\delta = \frac{8C_0}{(\gamma^2-1)(\alpha+3)}\int_0^1 (gf^2+h)\lambda^\alpha \mathrm{d}\lambda, \tag{69}$$

where $C_0 = 1, 2\pi, 4\pi$ for Cartesian, cylindrical, or spherical coordinates. The integral is singular at the lower limit. For details on its evaluation, see [17] or [18]. The variables f, g, and h in (69) are the non-dimensional velocity, density, and pressure behind the shock, while the integration is over the non-dimensional position, λ. Specifically,

$$r/r_2 = \lambda = x_1^{-\alpha_0} x_2^{\alpha_2} x_3^{-\alpha_1}, \tag{70}$$
$$u/u_2 = f = x_1\lambda, \tag{71}$$
$$\rho/\rho_2 = g = x_2^{\alpha_3} x_3^{\alpha_4} x_4^{\alpha_5}, \tag{72}$$
$$P/p_2 = h = x_3^{\alpha_4-2\alpha_1} x_4^{1+\alpha_5}. \tag{73}$$

The values immediately behind the shock can be found from the Rankine–Hugoniot conditions

$$\rho_2 = \frac{\gamma+1}{\gamma-1}\rho_1, \quad u_2 = \frac{2U}{\gamma+1}, \quad p_2 = \frac{2\rho_1 U^2}{\gamma+1}. \tag{74}$$

In (70) through (73), the exponents are given by

$$\begin{aligned}
\alpha_0 &= \frac{2}{\alpha+3},\\
\alpha_1 &= \frac{(\alpha+3)\gamma}{2+(\alpha+1)(\gamma-1)}\left[\frac{2(\alpha+1)(2-\gamma)}{\gamma(\alpha+3)^2} - \alpha_2\right],\\
\alpha_2 &= -\frac{\gamma-1}{2(\gamma-1)+\alpha+1},\\
\alpha_3 &= \frac{\alpha+1}{2(\gamma-1)+\alpha+1},\\
\alpha_4 &= \frac{\alpha+3}{2-\gamma}\alpha_1,\\
\alpha_5 &= -\frac{2}{2-\gamma}.
\end{aligned} \tag{75}$$

Recall that $\alpha = 0,\ 1,\ 2$ for Cartesian, cylindrical, or spherical coordinates, respectively. Given the constants

$$\begin{aligned}
a &= \frac{(\alpha+3)(\gamma+1)}{4},\\
b &= \frac{\gamma+1}{\gamma-1},\\
c &= \frac{\gamma(\alpha+3)}{2},\\
d &= \frac{(\alpha+3)(\gamma+1)}{(\alpha+3)(\gamma+1)-2\left[2+(\alpha+1)(\gamma-1)\right]},\\
e &= \frac{2+(\alpha+1)(\gamma-1)}{2},
\end{aligned} \tag{76}$$

the variables denoted by x_k are

$$\begin{aligned}
x_1 &= aV, & x_2 &= b(cV-1),\\
x_3 &= d(1-eV), & x_4 &= b(1-cV/\gamma).
\end{aligned} \tag{77}$$

In the equations for x_k, $V = tu/r$ is the similarity variable.

These expressions were used to find a solution such that $r_2 = r_\mathrm{f}$ at $t = 1$ using the following algorithm.

1. Find $\mathcal{E}$ using (67) evaluated at $t = 1$.
2. For each $t > 0$ find
   (a) The shock position from (67).
   (b) The shock speed from (68).
   (c) The post-shock variables from (74).
   (d) For each r, $0 \le r < r_2$, find
      (i) V using (70) and a root finding technique.
      (ii) The primitive variables using (71) through (73).

The algorithm is from [17] except the singular cases are not included, and the determination of $\mathcal{E}$ has been simplified. For the near vacuum conditions that develop near the origin, the algorithm requires very precise values for V. This was accomplished by using the Multiprecision Computing Toolbox for MATLAB [20].

8.2 Spherical test case

For the simulation, the length scale was taken to be the final shock position r_f, while the time scale was taken to be the corresponding time, t_f. The primitive variables were scaled as $\widetilde{\rho} = \rho/\rho_1$, $\widetilde{u} = u/(r_\mathrm{f}/t_\mathrm{f}) = u/u_r$, and $\widetilde{P} = P/(\rho_1 u_r^2)$. From [17] or equivalently [21], $\delta = 0.85106 = \widetilde{\mathcal{E}}$, where the appropriate scale for $\mathcal{E}$ is $\rho_1 u_r^2 r_\mathrm{f}^3$. Outside the first cell, the back pressure was set to $\widetilde{P}_1 = 1\times 10^{-12}$. Using this initial data, all three forms were run using a spatial increment of $\Delta\widetilde{r} = 0.01$ and a time step of $\Delta\widetilde{t} = 1\times 10^{-7}$. The time step was more than sufficient to ensure stability.

8.2.1 Accuracy

In Fig. 9, the primitive variables are plotted for the exact solution, and the numerical solutions using all three forms and AUSM$^+$(0,0) with $\Delta\widetilde{r} = 0.01$. The geometric form is clearly the closest to the correct values. The other two forms result in a shock in the incorrect location.

For the geometric form, there is a large error in the magnitude of the density just behind the shock. This is because the peak is sharp and because according to the exact solution (74), the density must instantaneously become six. This behavior of the point source is of course impossible to replicate numerically. Setting the density to six in the first cell only mildly improves the results. Increasing the spatial dis-

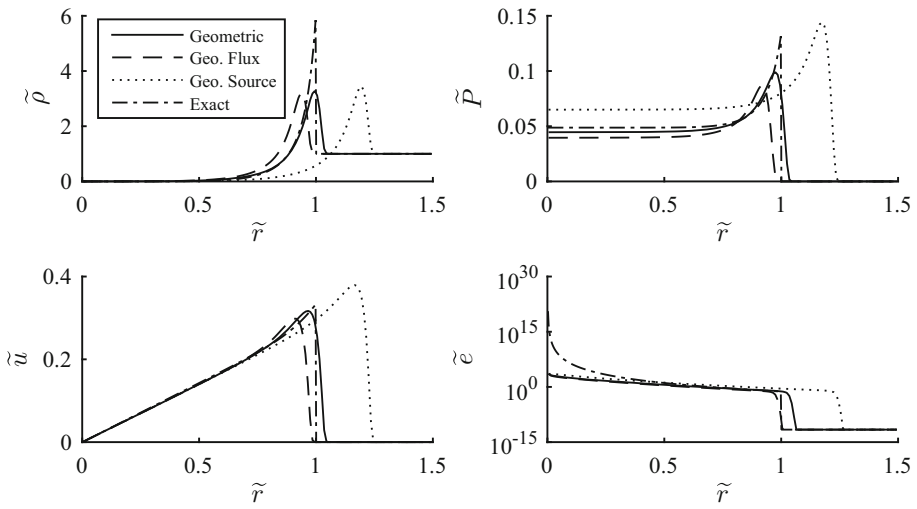


Fig. 9 Primitive variables for the Sedov solution in spherical coordinates using AUSM$^+$-up(0,0) and $\Delta\widetilde{r} = .01$

cretization or using MUSCL does reduce the error in $\widetilde{\rho}$ at $\widetilde{t} = 1$. However, at earlier times, the error is still quite large. The second-order results using the same discretization and initial conditions are given in Fig. 10. For all of the numerical methods, the post-shock density slowly builds toward six (and continues to get closer if the simulation is run longer). For the geometric form, the agreement between the pressure and velocity with the exact solution is quite good and improves if MUSCL or a finer discretization is used. The analytical solution for the internal energy becomes unbounded at the origin as time progresses, a feature which none of the methods come close to replicating. It should be noted that the internal energies were plotted on a log scale to show this discrepancy at the origin. However, the log scale distorts where it appears the shock location is for the various methods. The energy is found from the ideal gas law, so the actual shock location is as it appears in the density and pressure curves. As with the steady-state solution, using the center of volume as the point of evaluation had negligible effect on the end numerical results other than where the values in the first few cells were considered to be located.

The percent errors in the final shock locations as measured by the location of the peak value of the density functions are given in Table 4 for all three forms. For each form, the effect of the grid spacing was investigated by using $\Delta\widetilde{r} = 0.01$, 0.005, and 0.0025. In addition, runs were made with and without numerical diffusion and with and without MUSCL. The shock locations for the geometric and geometric flux form are essentially independent of the grid spacing, spatial order, and numerical diffusion. The geometric form gives nearly the exact location, while the geometric flux form underestimates the shock position by about 6% in all cases. The geometric source form considerably overestimates the shock position. Increasing the order of the method using MUSCL, without numerical diffusion, actually makes the final shock position for the geometric source form slightly worse. This may be due to the fact that most of the error occurs in the initial cell and the second-order interpolation is amplifying this error. Adding numerical diffusion makes the final shock location far worse for the geometric source form both with and without MUSCL. The predicted (Pred.) columns in Table 4 will be explained in the next subsection.

8.2.2 Conservation of energy and shock location

The misplacement of the shock was found to be directly related to the conservation of energy. Plots of the ratio of total energy in the domain over time to the original total energy for each method are given in Fig. 11 using $\Delta\widetilde{r} = 0.01$. As can be seen in the figure, only the geometric form is conservative. The total energy for the geometric flux form converges fairly quickly to its final value of approximately 75% of the original, while the total energy of the geometric source form at first sharply increases and then continues to increase, but at a much slower rate. To explore this further, the total energy in the first cell (using the geometric form) is also plotted in Fig. 11. Based on the theoretical discussions presented in Sect. 5 (stemming from the approximations in the derivations in Sect. 2), we can expect the majority of the error for the geometric flux and geometric source forms to occur as the energy is convected out of the first cell. Once the energy has left the first cell, the geometric flux form essentially conserves energy. In the case of the geometric source form, the error is the largest in the first cell, but also extends further.

These results are consistent with the analytical results in that the major error in using r instead of $\widehat{r}$ occurs near the origin. Both the geometric flux form and the geometric source form use this approximation. The incorrect volume represented by using r instead of $\widehat{r}$ is the only source of conservation error in the geometric flux form. In fact, the effect of dropping the $\Delta r^2/12$ term in the definition of $\widehat{r}$ is to reduce the volume of the first cell to 75% of its actual value,

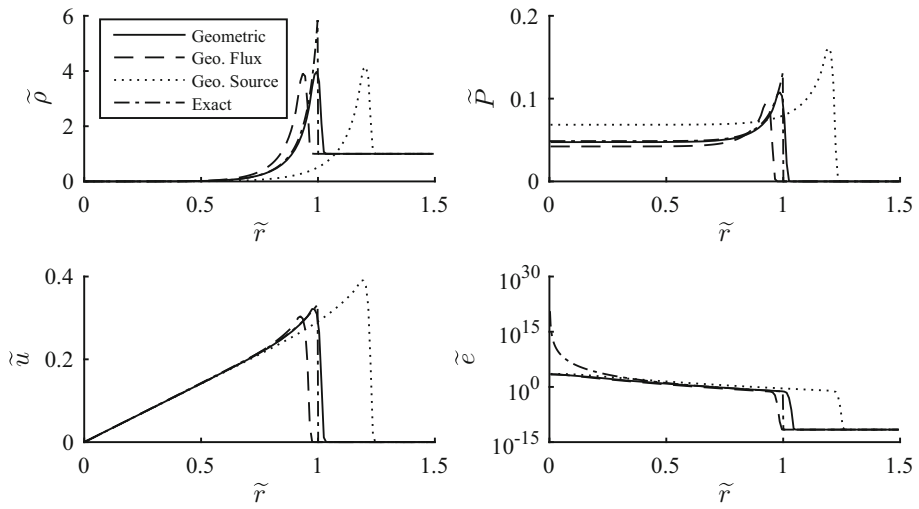


Fig. 10 Primitive variables for the Sedov solution in spherical coordinates using AUSM$^+$-up(0,0) and MUSCL

Table 4 Computed and predicted percent errors in shock location at $\widetilde{t} = 1$ for the Sedov solution in spherical coordinates

Form	$\Delta\widetilde{r}$	First order				Second order			
		AUSM$^+$(0,0)		AUSM$^+$(0.25,0.75)		AUSM$^+$(0,0)		AUSM$^+$(0.25,0.75)	
		Actual	Pred.	Actual	Pred.	Actual	Pred.	Actual	Pred.
Geometric	0.0100	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.0050	− 0.50	0.00	− 0.50	0.00	0.00	0.00	0.00	0.00
	0.0025	− 0.50	0.00	− 0.50	0.00	− 0.25	0.00	− 0.25	0.00
Geometric flux	0.0100	− 6.03	− 5.59	− 6.03	− 5.59	− 6.03	− 5.59	− 6.03	− 5.59
	0.0050	− 6.01	− 5.59	− 6.01	− 5.59	− 6.01	− 5.59	− 6.01	− 5.59
	0.0025	− 6.01	− 5.59	− 6.01	− 5.59	− 5.76	− 5.59	− 5.76	− 5.59
Geometric source	0.0100	20.10	19.84	33.17	33.10	21.11	20.89	29.15	28.71
	0.0050	19.55	20.03	33.08	33.27	21.05	21.19	29.07	29.03
	0.0025	19.77	20.12	33.04	33.36	21.28	21.37	29.04	29.21

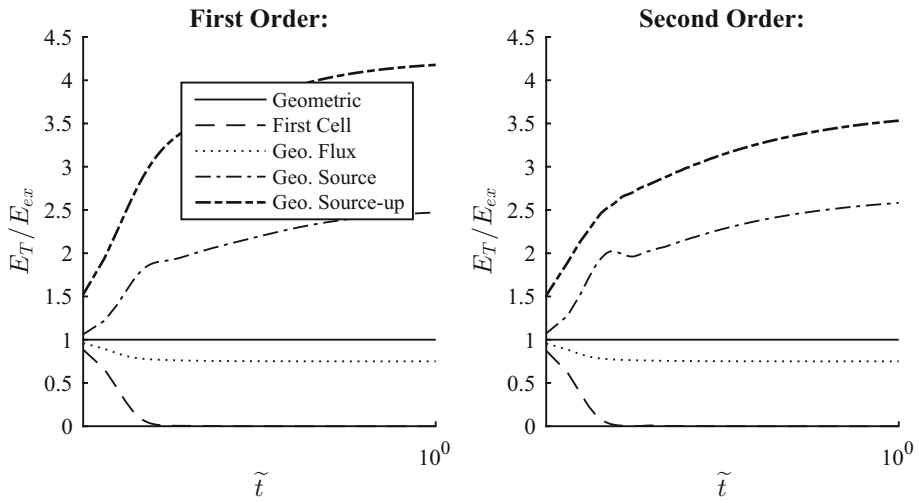


Fig. 11 Total energy in entire domain and in the first cell with $\Delta\widetilde{r} = 0.01$ for the Sedov solution in spherical coordinates. All results are using AUSM$^+$-up(0,0) except those denoted by "Geo. Source-up" which use AUSM$^+$-up(0.25,0.75). Second-order results are with MUSCL. Note the $\widetilde{t}$-axis is log scale

which is precisely the percent of the total initial energy the results converge toward. The geometric source form, also accrues conservation error through treating part of the fluxes as sources of $\mathcal{O}(1/r)$. This effect in the first cell is apparently larger than the reduction caused by the volume approximation and results in an approximate doubling of the total energy by the time the first cell is evacuated. By (56), the error from emptying the first cell is proportional to the difference in the interfacial flux out of the right of the cell and the same quantity evaluated at the center of the cell. Thus, the error accrued as the first cell is emptied will depend on the method used to find the interfacial fluxes and the properties of the flow. In particular, it can be expected to change when numerical diffusion and/or MUSCL are used.

Converting to second order or using a finer discretization serves to hasten the emptying of the first cell. This has little impact on energy conservation when either the geometric or geometric flux form is used. Without numerical diffusion, both a finer discretization and using MUSCL actually make energy conservation slightly worse for the geometric source form. As was shown in Sect. 5, the conservation error of the geometric form is independent of the order of the method, while in general the conservation error of the geometric source form is not. Its dependence on the value of the interfacial flux with respect to the same quantity evaluated at the cell center can make it a function of the spatial order and even the details of the method that is used to obtain second order. A smaller $\Delta\widetilde{r}$ either not changing the conservation error for the geometric flux form or making it worse for the geometric source form may seem to contradict the results of Sect. 5. However, all of the energy is initially in the first cell; thus, N in (56) and (51) is in effect one as the first cell is quickly evacuated.

Adding numerical diffusion also has negligible effect on the conservation error for the geometric and geometric flux form. In contrast, the conservation error for the geometric source form becomes considerably worse. For the first-order results, by $\widetilde{t} = 1$, there is over four times as much energy in the system as there was at the beginning. Using MUSCL improves the results, but there is still approximately 3.5 times as much energy at $\widetilde{t} = 1$ (see Fig. 11). The curves for the geometric form and geometric flux form are so close to the curves without numerical diffusion that they have been omitted to avoid clutter.

For all of the forms, discretizations, and methods, it was discovered that using the erroneous final energies in (67) (with $\widetilde{t} = 1$) predicts the computed shock locations given in Table 4 within 0.5%. These values are given in the predicted (Pred.) columns in the table. Hence, the underestimate of the shock location by the geometric flux form and the large overestimate of the shock location by the geometric source form are directly related to total energy conservation. Therefore, a more precise way to interpret the shock locations at $t = 1$ is that they are essentially correct for the amount of energy that is emitted from the first cell for the geometric flux form and the final total energy for the geometric source form.

8.2.3 Truncation error

Having an exact solution provides another opportunity to check the truncation errors of the different forms. Unlike the steady-state solutions, the velocities are supersonic for a portion of the domain. To avoid the difficulties modeling the singularity at $\widetilde{t} = 0$ encountered in the previous section, the solution was initialized to the values of the analytical solution when the shock is at $\widetilde{r} = 1$ and $\widetilde{t} = 1$. The domain was divided into cells with centers at $\widetilde{r}_i = 0.0225 + i\,\Delta\widetilde{r}$, with $\Delta\widetilde{r} = 0.005, 0.01$. This provided cells with common midpoints for error comparison and space for up to two ghost cells for the second-order MUSCL scheme to the left of the numerical domain. The region was not discretized all of the way to the origin because the energy becomes unbounded there.

Using MUSCL to convert to second-order spatial convergence brought out some consequences as to which variables are interpolated. It is recommended in [10] that (ρ, u, H) be interpolated and pressure be found from the closure equations. However, given the large growth in the energy near the origin, it was found that interpolating H and ρ linearly resulted in a nonlinear interpolation of the pressure via the ideal gas conditions. This led to an apparent disruption of the pressure balance in the momentum equation for both forms and unacceptably large relative errors in velocity (see Fig. 12). Two other options were investigated: interpolating (ρ, u, H, P) and interpolating (ρ, u, P). Both had a similar end result in terms of error, and both were quite superior to interpolating (ρ, u, H). Since interpolating (ρ, u, P) still used the ideal gas equations at the interface, it was chosen.

The absolute relative errors in pressure and density for the finest discretization are plotted in Fig. 13. The absolute relative errors in internal energy and density are very similar qualitatively and quantitatively. The two discretizations were used to verify the convergence orders are near one without MUSCL and near two when MUSCL is used. Unlike the steady case, the geometric source form has slightly smaller relative errors for both the spatially first- and second-order results. However, as was seen in the previous subsections, if the geometric source form was used starting at $t = 0$, the shock would not even be in the correct location. Although the shock locations were not changed appreciably for the geometric and geometric flux forms by changing to second-order, the truncation errors are superior by a few orders of magnitude when MUSCL is used. The kinks in the graphs around $\widetilde{r} = 0.9$ are from flow transitions to the supersonic regime. The spikes in the relative error in pressure curves in the middle of the range are due to the numerical approximations passing through the exact solution. Numerical diffusion

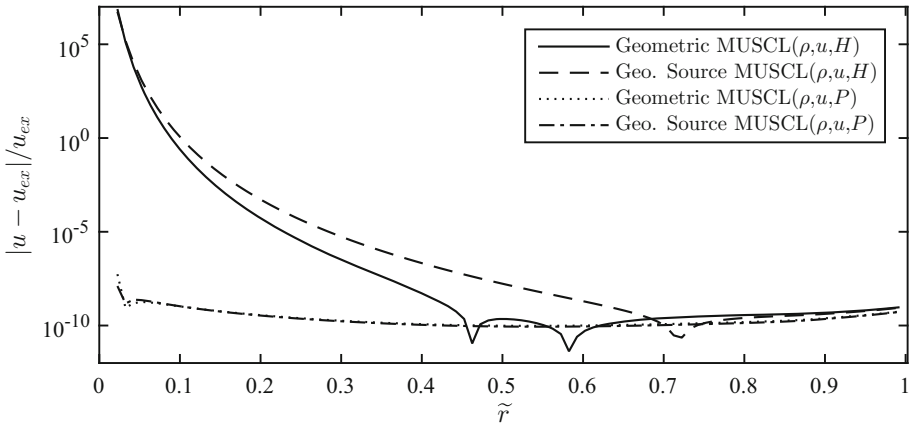


Fig. 12 Absolute relative error in velocity for Sedov solution in spherical coordinates when (ρ, u, H) and (ρ, u, P) are interpolated using MUSCL with $\Delta\widetilde{r} = 0.005$

increases the errors slightly but has been omitted from the figure to avoid clutter.

8.3 Cylindrical test case

For a cylindrical test case, the same discretizations and scalings were used as for the spherical case. For the initial conditions, in cylindrical coordinates $\delta = 0.984041$ [17]. As with the spherical case, numerical conservation and shock location appear to be directly related. The percent error in the numerically estimated shock locations and those predicted by the final energy contained in the solutions are summarized in Table 5. Recall in the cylindrical case, the geometric and geometric flux forms are identical. The conservation errors and resulting shock locations of the geometric source form follows a similar pattern as with spherical coordinates. However, with only the treatment of the source terms causing the errors, they are smaller (see Fig. 14). The truncation errors follow the same pattern as the spherical case, but the errors are also much smaller in magnitude. As with the spherical case, it was also found to be best to interpolate ρ, u, and P rather than ρ, u, and H when using MUSCL.

9 Shock tubes

Spherical and cylindrical shock tube problems were used to evaluate the geometric and geometric source forms with respect to conservation, accuracy, and feature location. In the interest of brevity, the geometric flux form was omitted from the comparisons. An initial high-pressure region, P_h, is assumed for $r \leq r_0$ along with an ambient pressure of P_a for $r > r_0$. The initial temperature is assumed constant, and density is found from the ideal gas properties of air ($\gamma = 1.4$). The primitive variables are scaled as $\widetilde{\rho} = \rho/\rho_\mathrm{a}$, $\widetilde{P} = P/(\rho_\mathrm{a} a_\mathrm{a}^2)$, $\widetilde{T} = T/T_\mathrm{a}$, and $\widetilde{u} = u/a_\mathrm{a}$, where a_a is the speed of sound in the ambient region. The spatial scale is r_0, and the time scale is r_0/a_a. Using these scalings and initial conditions, it is only necessary to specify the pressure ratio and γ. The non-dimensional ideal gas constant is $1/\gamma$. The pressure ratio between the high- and low-pressure region is $P_\mathrm{h}/P_\mathrm{a} = 40{,}000$ for the spherical shock tube and 900 for the cylindrical shock tube.

9.1 Spherical shock tube

The primitive variables over time for the spherical shock tube are plotted during the initial expansion in Fig. 15. Initially, a spike in the velocity develops at the edge of the high-pressure region and a main shock front is formed. Behind the main shock, a contact discontinuity and a secondary shock are formed as an expansion fan travels back toward the origin. For example, in Fig. 15 in the curves for $\tilde{t} = 10$, the primary and secondary shocks are easily seen in the pressure and velocity curves and are near $\tilde{r} = 19.2$ and $\tilde{r} = 24.2$, respectively. For the same time, the contact discontinuity can be seen either in the density curve, just past the maximum, or in the temperature curve just prior to its maximum. At $\tilde{t} = 10$, it is at about $\tilde{r} = 19.9$. During the initial time period, approximately $\tilde{t} \in [0, 30]$, all of these features decrease in magnitude and travel to the right at a decreasing rate due to both the geometric expansion and the development of near vacuum conditions near the origin.

In Fig. 16, the locations of the primary and secondary shocks and the contact interface are plotted as a function time. The primary shock continues to propagate outwards, while eventually the decrease in pressure causes the inner flow to begin to collapse. The secondary shock and contact interface reverse direction and accelerate toward the origin. The secondary shock reflects off of the origin and then collides with the contact interface. A portion of the secondary shock is reflected back toward the origin as a tertiary wave. This feature is not plotted and increasing the discretization greatly effects its location. The secondary shock passing through the contact discontinuity reverses the of travel of the contact to also being outward. The

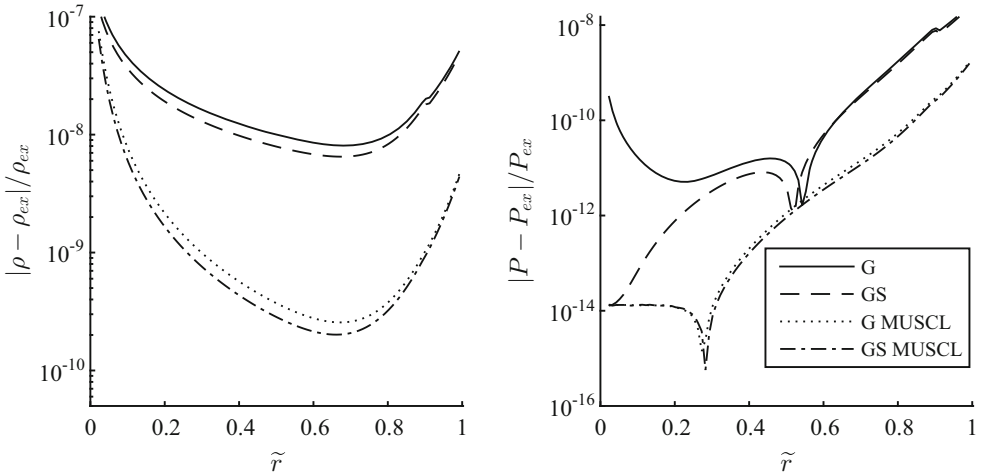


Fig. 13 Absolute relative error in density and pressure for the Sedov solution in spherical coordinates after one time step using AUSM$^+$(0,0) with the geometric (G) and geometric source forms (GS)

Table 5 Computed and predicted percent errors in shock location at $\widetilde{t} = 1$ for Sedov solution in cylindrical coordinates

Form	$\Delta\widetilde{r}$	First order				Second order			
		AUSM$^+$(0,0)		AUSM$^+$(0.25,0.75)		AUSM$^+$(0,0)		AUSM$^+$(0.25,0.75)	
		Actual	Pred.	Actual	Pred.	Actual	Pred.	Actual	Pred.
Geometric	0.0100	− 1.00	0.00	− 1.00	0.00	− 1.00	0.00	− 1.00	0.00
	0.0050	− 1.00	0.00	− 1.00	0.00	− 0.50	0.00	− 0.50	0.00
	0.0025	− 0.75	0.00	− 0.75	0.00	− 0.25	0.00	− 0.25	0.00
Geometric source	0.0100	6.03	6.98	14.85	14.07	9.05	9.25	13.45	13.06
	0.0050	6.02	7.04	14.90	14.04	9.02	9.39	13.58	13.03
	0.0025	6.26	7.07	14.92	14.27	9.26	9.47	13.66	13.27

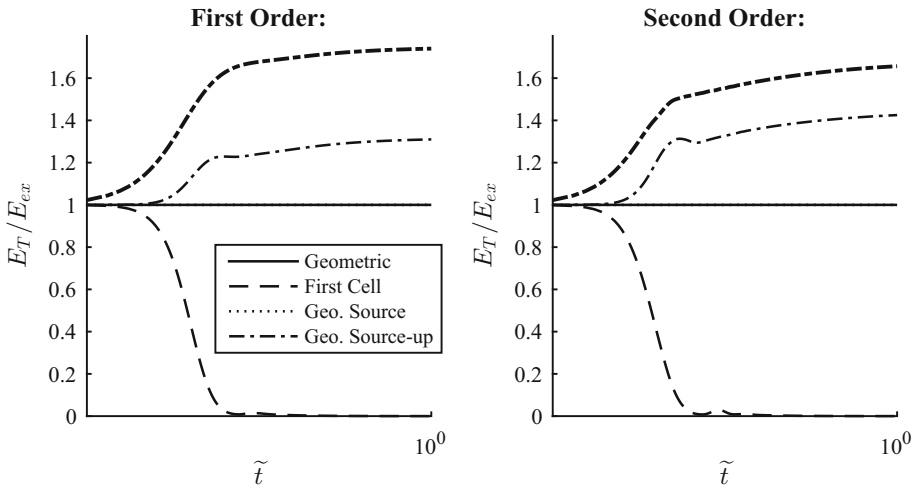


Fig. 14 Total energy in entire domain and in the first cell with $\Delta\widetilde{r} = 0.01$ for the Sedov solution in cylindrical coordinates. All results are using AUSM$^+$-up(0,0) except those denoted by "Geo. Source-up" which use AUSM$^+$-up(0.25,0.75). Second-order results are with MUSCL. Note the $\widetilde{t}$-axis is log scale

region around the origin again becomes evacuated. Eventually, the contact interface once again reverses direction and moves toward the origin. Qualitatively, this overall progression of the flow matches that described by Brode's [22,23] simulation of the detonation of a spherical charge of TNT. The details of determining the feature locations plotted in Fig. 16 will be discussed in the next subsection

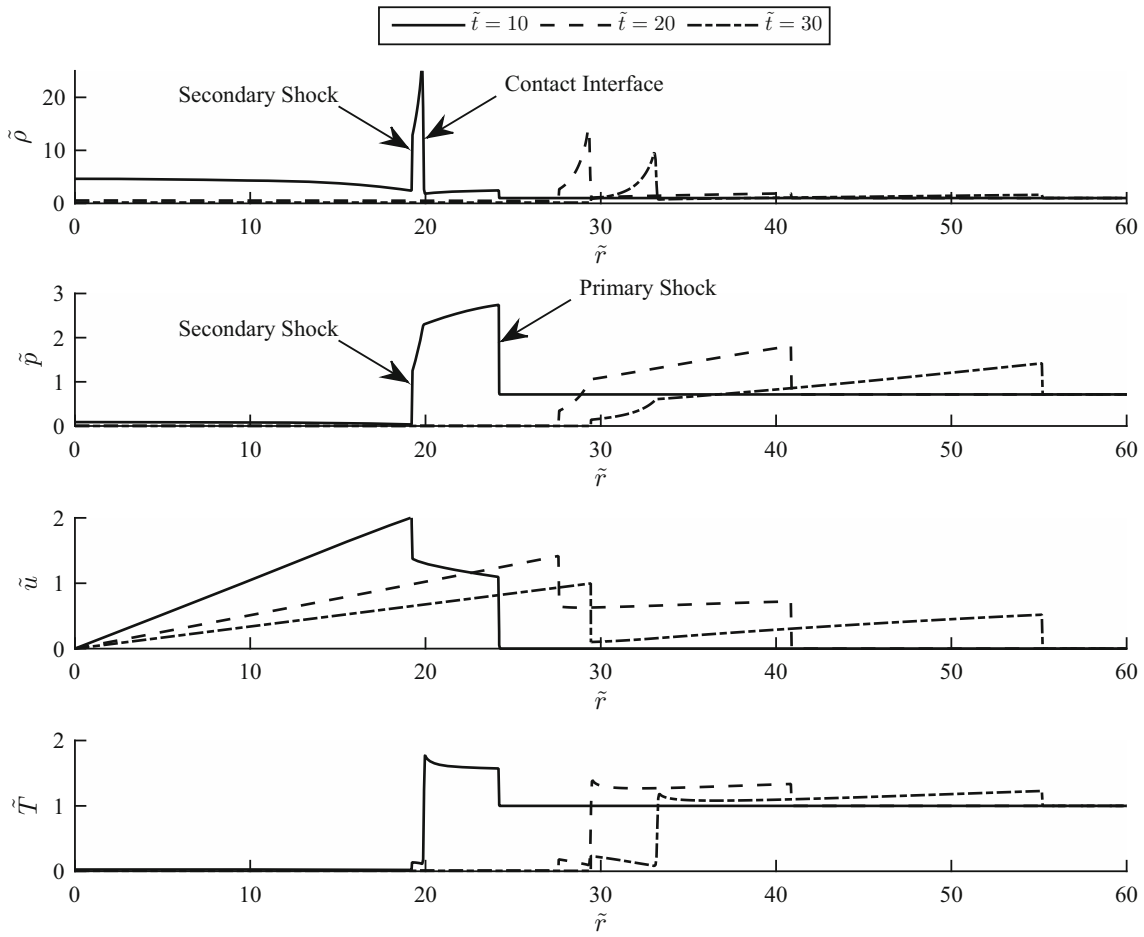


Fig. 15 Initial expansion of high-pressure region of spherical shock tube. Primary shock wave, contact discontinuity, and secondary shock wave travel outwards

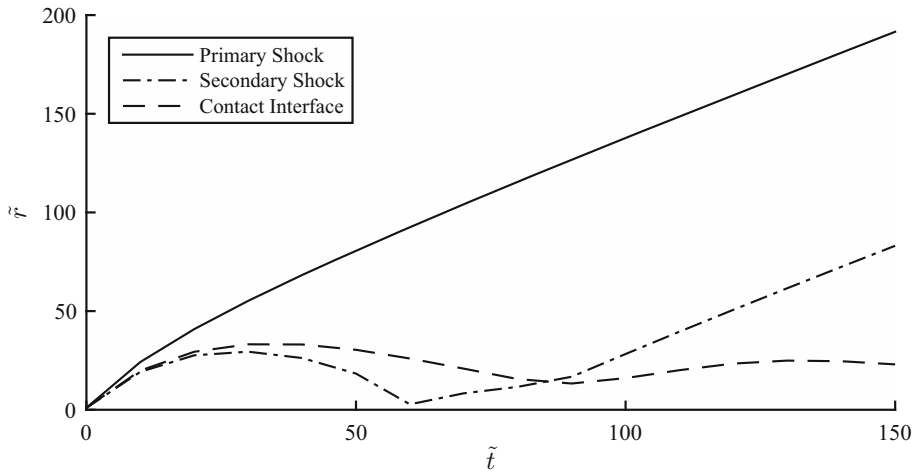


Fig. 16 Feature locations using geometric form for a spherical shock tube with $\Delta\tilde{r} = 0.0025$, AUSM$^+$-up(0.25,0.75), and MUSCL

9.1.1 Locating features

The radial location of the shocks and contact interface vary depending on which form and order is used. Based upon previous results, it is assumed that the geometric form is most accurate. Both the shocks and the contact interface theoretically are perfect discontinuities in the primitive variables, but of course smeared over a few grid cells in a numerical method. Various approaches could be taken as to determining the location of a feature based on the numerical approxi-

mations. In this section, the discontinuities are assumed to occur at local extrema of the slope (or points of inflection) as is done in [24]. Therefore, numerically, the shocks (or contact discontinuity) were taken to be at a point of maximum derivative as found by interpolating between the two largest values to the point where the second derivative would be zero. Since the points of evaluation for the geometric form are unevenly spaced, the following second-order approximations were used for the first and second derivative [15]:

$$w'(r_{\mathrm{v},j}) \approx \frac{\Delta r_{\mathrm{v},j-1} w_{\mathrm{v},j+1}}{\Delta r_{\mathrm{v},j}\left(\Delta r_{\mathrm{v},j}+\Delta r_{\mathrm{v},j-1}\right)} + \frac{\left(\Delta r_{\mathrm{v},j}-\Delta r_{\mathrm{v},j-1}\right) w_{\mathrm{v},j}}{\Delta r_{\mathrm{v},j-1}\Delta r_{\mathrm{v},j}} - \frac{\Delta r_{\mathrm{v},j} w_{\mathrm{v},j-1}}{\Delta r_{\mathrm{v},j-1}\left(\Delta r_{\mathrm{v},j}+\Delta r_{\mathrm{v},j-1}\right)}, \tag{78}$$

$$w''(r_{\mathrm{v},j}) \approx \frac{2 w_{\mathrm{v},j+1}}{\Delta r_{\mathrm{v},j}\left(\Delta r_{\mathrm{v},j}+\Delta r_{\mathrm{v},j-1}\right)} - \frac{2 w_{\mathrm{v},j}}{\Delta r_{\mathrm{v},j}\Delta r_{\mathrm{v},j-1}} + \frac{2 w_{\mathrm{v},j-1}}{\Delta r_{\mathrm{v},j-1}\left(\Delta r_{\mathrm{v},j}+\Delta r_{\mathrm{v},j-1}\right)}. \tag{79}$$

When the cell centers are used for the geometric source form instead of the centers of volume, both approximations simplify to the standard centered difference formulas.

For the primary and secondary shock, the local absolute maximum in the pressure derivative was used, while for the contact interface, the derivative of density was used prior to the reflection of the secondary wave since it was sharper, while temperature was used afterward since as the secondary shock collided with the density peak near the contact interface the two nearby discontinuities in density became unclear. Using other variables, such as velocity to determine the primary shock location, does change the feature locations slightly. The locations of the primary shock, contact interface, and secondary shock for the geometric form with a very small discretization ($\Delta\tilde{r} = 0.0025$) and MUSCL are plotted in Fig. 16.

9.1.2 Error in feature locations over time

In Fig. 17, the absolute errors in the feature locations are plotted over time treating the second-order geometric form with $\Delta\tilde{r} = 0.0025$ as exact. Separate error curves are plotted for three spatial discretizations, $\Delta\tilde{r} = 0.04,\ 0.02,\ 0.01$, and with and without MUSCL. The first-order results are using AUSM$^+$(0,0), while for the second-order results AUSM$^+$(0.25,0.75) is used. Numerical diffusion was necessary to avoid oscillations upon reflection of the secondary shock.

For the primary shock, the first-order geometric form results are comparable to all but the finest discretization of the second-order geometric source form and for early times are as accurate as the second-order geometric form with twice the grid spacing.

Initially, the first- and second-order geometric source form tends to overestimate the secondary shock locations, while the first-order geometric form tends to underestimate the location. When the flow reverses direction, since the shocks for the geometric form are closer to the origin, they are reflected first and become overestimates for the position, while the first-order geometric source form results become underestimates. Colliding with the contact discontinuity around $\tilde{t} = 80$ seems to temporarily effect the ordering of the predictions of the positions of the secondary shock (see Fig. 17), but by $\tilde{t} = 90$ they are back in the order that existed just after the reflection.

The position of the contact interface prior to the reflection of and interaction with the secondary shock is predicted well by the first-order geometric form. After the reflection of the secondary shock, the accuracy in the location of the contact interface appears to only be dependent upon the order of the method and the discretization size and not the form. This may in part be due to switching from using the density to temperature to detect the contact interface after the reflection.

9.1.3 Root-mean-square error in feature location

The rms errors of the feature locations using the "exact" solution are given in Table 6. The primary shock location for the spatially second-order results is over twice as accurate by this measure when the geometric form is used versus the geometric source form. The first-order geometric form primary shock locations are even comparable to the second-order geometric source form results. Alternately, the geometric form results for the primary shock location are more (or very close to as) accurate as the geometric source form with half the grid spacing. For the secondary shock, the locations using the geometric form without MUSCL are around twice as accurate as the first-order geometric source form. The advantage for the geometric form in predicting the location of the secondary shock is less pronounced for the second-order results. The location of the contact interface has the largest errors of any of the features, and the accuracy is primarily a function of the order and discretization level rather than the form. However, for all of the feature locations, if the order and discretization level are equal, the geometric form is at least slightly superior to the geometric source form.

The differences are probably the most pronounced for the primary shock because initially most of the mass and energy and the conservation errors associated with it are concentrated near the origin and travel outward with the primary shock. Additionally, the reflection of the secondary shock

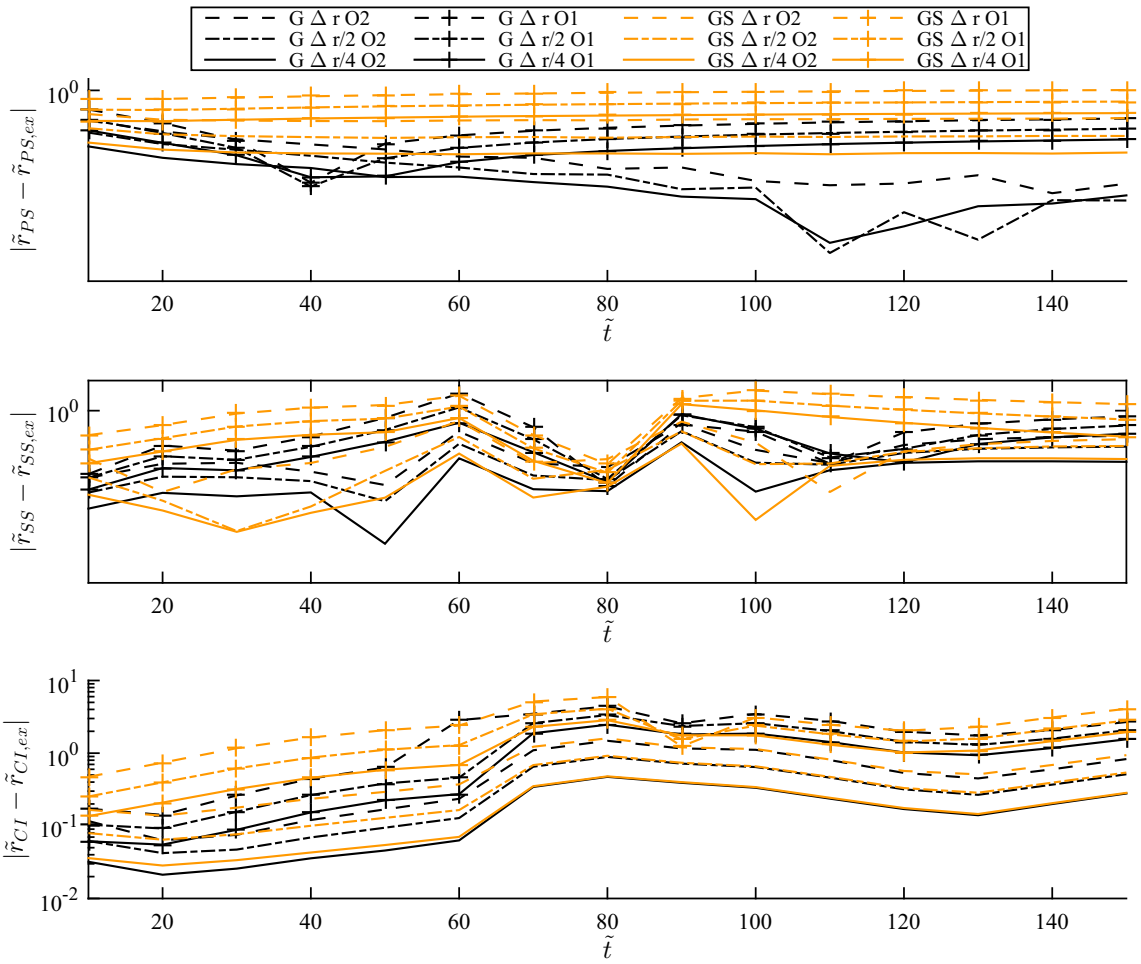


Fig. 17 Absolute error in feature locations using the geometric (G) and geometric source forms (GS). The order one (O1) results are with AUSM$^+$(0,0), while the order two (O2) results are with AUSM$^+$-up(0.25,0.75) and MUSCL. The largest spatial increment is $\Delta\tilde{r} = 0.04$

Table 6 Root-mean-square error in feature location for spherical shock tube problem treating AUSM$^+$-up(0.25,0.75), $\Delta\tilde{r} = 0.0025$, with MUSCL as exact

$\Delta\tilde{r}$	Geometric form			Geometric source form		
	0.04	0.02	0.01	0.04	0.02	0.01
Primary shock: Order 2	5.11e−02	2.48e−02	1.08e−02	1.77e−01	6.51e−02	2.46e−02
Order 1	1.45e−01	7.75e−02	4.01e−02	8.47e−01	4.27e−01	2.15e−01
Secondary shock: Order 2	1.99e−01	1.21e−01	6.30e−02	2.37e−01	1.31e−01	6.64e−02
Order 1	7.58e−01	4.35e−01	3.00e−01	1.62e + 00	9.26e−01	5.68e−01
Contact interface: Order 2	7.43e−01	4.45e−01	2.36e−01	8.07e−01	4.64e−01	2.42e−01
Order 1	2.39e + 00	1.74e + 00	1.27e + 00	2.94e + 00	2.02e + 00	1.43e + 00

and the relatively slow movement of the contact interface and its reversals in direction of travel serve to diminish the differences in the errors in predicting their positions. Most of the variation in the locations of the contact interface that make its location errors larger than the other features occur as it interacts with the secondary shock, making it less sharp.

9.1.4 Numerical conservation

The relative error in conservation in excess mass is plotted for both forms in Fig. 18 with $\Delta\tilde{r} = 0.04$, 0.02, and 0.01 using both the first-order AUSM$^+$(0,0) and AUSM$^+$(0.25,0.75) with MUSCL. Since the size and number of the time steps is essentially proportional to the grid spacing, Courant numbers

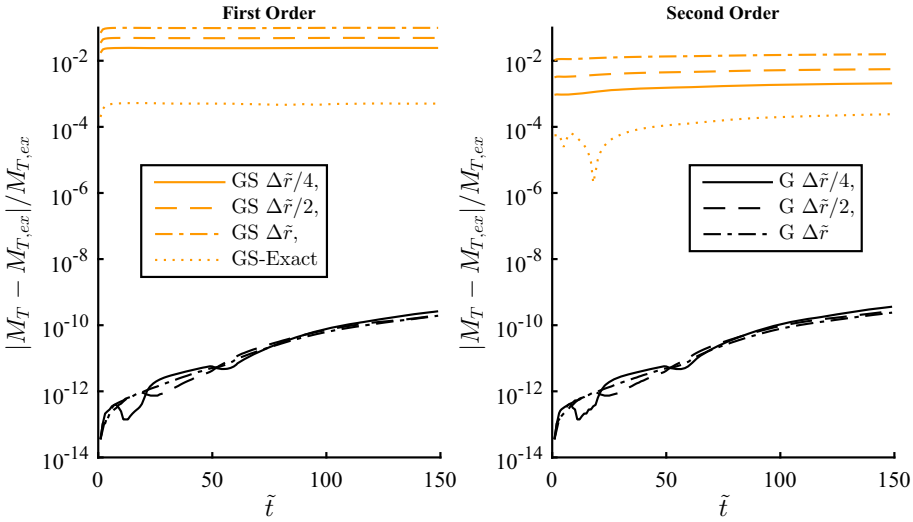


Fig. 18 Absolute relative error in excess mass in spherical shock tube for first- and second-order spatial results with initial grid spacing of $\Delta\tilde{r} = 0.04$ for the geometric (G) and geometric source (GS) forms. The geometric source form error level is converging toward "GS-Exact". First-order results are AUSM$^+$(0,0) and second-order results are AUSM$^+$(0.25,0.75) with MUSCL. Both legends apply to both graphs

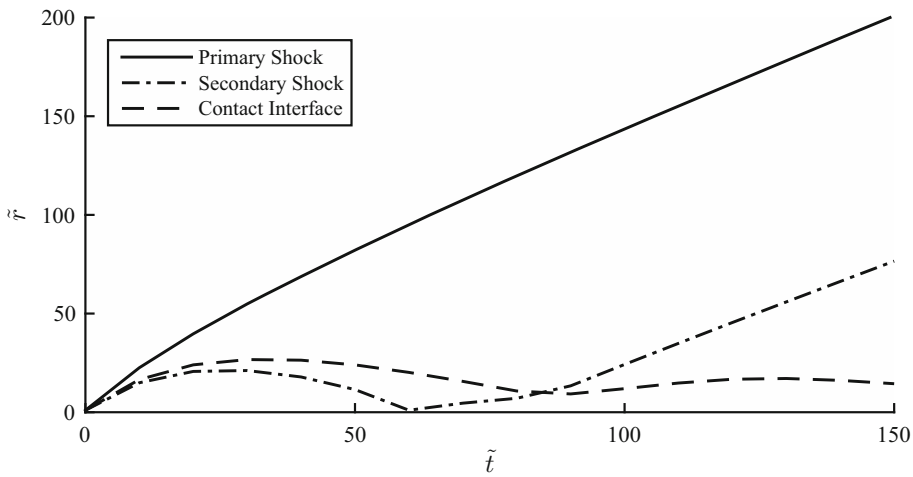


Fig. 19 Feature locations using geometric form for a cylindrical shock tube, $\Delta\tilde{r} = 0.0025$, AUSM$^+$-up(0.25,0.75), and MUSCL

Table 7 Root-mean-square error in feature location for cylindrical shock tube problem treating AUSM$^+$-up(0.25,0.75), $\Delta\tilde{r} = 0.0025$, with MUSCL as exact

$\Delta\tilde{r}$	Geometric form			Geometric source form		
	0.04	0.02	0.01	0.04	0.02	0.01
Primary shock: Order 2	4.29e−02	2.11e−02	9.42e−03	6.81e−02	2.65e−02	1.14e−02
Order 1	2.67e−01	1.51e−01	8.44e−02	1.34e−01	5.36e−02	2.17e−02
Secondary shock: Order 2	3.84e−01	5.20e−02	2.20e−02	4.72e−01	4.25e−02	2.02e−02
Order 1	4.56e−01	2.83e−01	1.75e−01	7.09e−01	3.14e−01	1.90e−01
Contact interface: Order 2	5.02e−01	2.94e−01	1.53e−01	5.39e−01	3.08e−01	1.58e−01
Order 1	1.94e + 00	1.44e + 00	1.04e + 00	2.20e + 00	1.56e + 00	1.10e + 00

of 0.125, 0.25, and 0.5 were used corresponding to the spatial increments so that the results would not be effected by differing numbers of time steps. The excess mass was defined to be that in excess of the amount that would have been present if the entire region had been at ambient conditions. The error in excess mass was used so the results would not be too skewed by the mass in the numerical domain that was either yet to be or never disturbed. There might be a better definition to use since the region around the origin is evacuated and therefore some of the base mass is in motion. In addition, for the geometric source form, the level of conservation the solutions are converging to was calculated using extrapolation (and is labeled "GS-Exact"). Numerically, the orders of convergence for the geometric source form are approximately 1.0 and 1.6 for the first- and second-order results, respectively, and the error level is not converging to zero. As can be seen in the

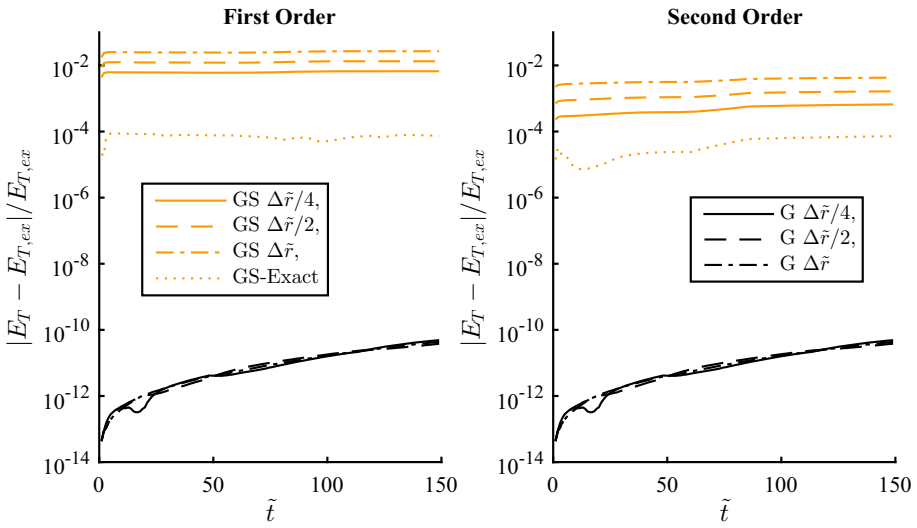


Fig. 20 Absolute relative error in excess energy in cylindrical shock tube for first- and second-order spatial results with initial grid spacing of $\Delta\tilde{r} = 0.04$ for the geometric (G) and geometric source (GS) forms. The geometric source form error level is converging toward "GS-Exact". First-order results are AUSM$^+$(0,0) and second-order results are AUSM$^+$(0.25,0.75) with MUSCL. Both legends apply to both graphs

figure, the majority of the error occurs initially when there is the largest amount of excess mass near the origin. The errors using the geometric form are several orders of magnitude better and appear to be largely independent of the order or step-size, as is predicted in Sect. 5.1.

9.2 Cylindrical shock tube

The initial conditions and discretizations are the same as for the spherical shock tube except that the pressure ratio is 900:1. The pressure ratio was lowered to obtain primary shock velocities of approximately the same magnitude as the spherical case. The locations of the primary shock, contact interface, and secondary shock for the geometric form with $\Delta\tilde{r} = 0.0025$ and MUSCL are plotted in Fig. 19. The contact interface and secondary shock wave do not travel quite as far from the origin as they did in the spherical test case.

The rms errors in the feature locations for first- and second-order results and $\Delta\tilde{r} = 0.04, 0.02, 0.01$ are given in Table 7 treating the discretization with $\Delta\tilde{r} = 0.0025$, the geometric form, and MUSCL as exact. As with the spherical case, the advantage of the geometric form is the largest for the primary shock location. However, without the additional error introduced by approximating the volume of the cells in spherical coordinates, the differences are somewhat smaller. The accuracy of the locations of the secondary shock and contact interface are primarily a function of the discretization level and form. In fact, the second-order geometric source form with the two highest discretizations has a smaller error than the geometric form for the secondary shock location. For the remainder of the results, the geometric form with the same discretization level and order is superior to the geometric source form. As with the spherical case, the errors in determining the contact interface location are the largest and primarily due to variations as it interacts with the secondary shock. The plots of the error over time have been omitted for brevity.

The error in the conservation of excess energy for both spatial orders and all discretizations is plotted in Fig. 20 along with the error level the geometric source form is converging toward. The results for mass are qualitatively and numerically similar. The results are also qualitatively similar to the spherical case, but smaller in magnitude. Again, numerically, the orders of convergence for the geometric source form is approximately 1.0 and 1.6 for the first- and second-order results, respectively. The errors in the geometric form are several orders of magnitude better and essentially independent of grid spacing or spatial order.

10 Conclusions

In this paper, the numerical properties of the geometric form, the geometric flux form, and the geometric source form are investigated analytically and numerically for radial compressible flow. The following is a list of the analytical contributions of this paper.

– Although they can be found elsewhere, derivations of discrete versions of all three forms are given along with clear geometric interpretations of the forms and their differences. These differences include the approximation of the control volume in spherical coordinates, the treatment of the fluxes, and the point of evaluation. The geometric source form and geometric flux form in spherical coordinates contain an approximation of the volume of each cell. The geometric source form treats the fluxes as Cartesian and then adjusts for the particular geometry using source terms. The primitive variables of the geometric

and geometric flux forms are evaluated at the center of volume of each cell, rather than the midpoint.
- Bounds on the numerical conservation error of all three forms are found analytically. The geometric form is numerically conservative independent of the flux scheme, grid size, or spatial order. Except in very special circumstances, the geometric flux form's numerical conservation error is $\mathcal{O}[\log(\Delta r)\Delta r^2]$, compared to $\mathcal{O}(\Delta r)$ for the geometric source form. The results can only depend on the flux scheme or its spatial order when the geometric flux form or the geometric source form are used. However, only the conservation rate constant, not the conservation order, is effected by the flux scheme.
- All three forms are found to result in constant total enthalpy for steady flows.
- An easily used form of the steady-state solution for radial flows is derived as a test problem for compressible flow.

Numerically, the steady-state solution, Sedov's similarity solution, and shock tube problems were used to verify the analytical results and further investigate each form. The AUSM$^+$-up method and its variations including numerical diffusion and its extension to second order using MUSCL were used to determine the fluxes. The findings based upon the numerical tests are as follows:

- A clear prescription for a conservative numerical method using AUSM$^+$-up and MUSCL appropriate for radial coordinates is given.
- The numerical conservation bounds are verified using all three test problems. The geometric and geometric flux form are many orders of magnitude better than the geometric source form by this measure.
- The geometric form is superior to the geometric source form with regard to truncation error for the steady-state solution.
- The constant enthalpy property for steady-state problems is obtained by the geometric and geometric source forms through different combinations of ρ, u, and P.
- The rms errors of the steady-state solution are smaller for the geometric form versus the geometric source form.
- The geometric form is far superior to the geometric source form and geometric flux form with regard to shock location for the Sedov problems. The initial concentration of energy at the origin emphasized the numerical conservation error caused by the volume approximation of the geometric flux form and the treatment of portions of the fluxes as sources by the geometric source form.
- A direct connection between the conservation errors of the forms and the final shock locations is found for the Sedov test problem. The strong shock location can be predicted by using the final total energy in the domain as the initial energy. For the geometric flux form, the error is primarily due to the approximation of the volume of the first cell, while the error caused by the treatment of the fluxes as sources dominates the numerical conservation error and shock location of the geometric source form.
- Based on the truncation errors for the Sedov problem, it is recommended that density, pressure, and velocity be interpolated rather than density, pressure, and enthalpy in order to achieve second order. This might not be generally applicable and could be investigated further in future work.
- A simple algorithm is outlined for determining compressible flow feature locations.
- The primary shock for the spherical and cylindrical shock tube problems is best located by the geometric form. The same is true for the secondary shock in spherical coordinates and without MUSCL in cylindrical coordinates. The contact interface is found most precisely by the geometric form, but the estimate of its position has the largest error of all of the features mainly due to its interaction with the secondary shock. In general, for the secondary shock and contact interface, the order and discretization level of the method are more important than the form of the equations.
- In general, the geometric source form appears to be most negatively impacted by numerical diffusion both with respect to numerical conservation and accuracy. This is probably because part of the fluxes are in the source terms and as a result are not diffused by the flux scheme.
- In general, the differences between the forms in cylindrical coordinates, which do not include an approximation of the volume of the cells, are not as pronounced.
- The point of evaluation, at least for the test problems considered, had only a minor effect on the end results.

Overall, the geometric form is recommended for problems with radial symmetry along with evaluation of the primitive variables at the centers of volume and interpolating density, velocity, and pressure to achieve second-order spatial results. Numerical diffusion tends to increase errors and should only be used when necessary. If the origin is not included in the domain, the somewhat simpler geometric flux form could be used and/or evaluation at the midpoint of the cell. The chief contribution of this paper is that it clearly shows that the conservation error in cylindrical and spherical coordinates depends on the form of the equations to which a flux method is applied and that this error is concentrated in the vicinity of the origin. For future efforts, different methods of enforcing the boundary conditions should be investigated as well as the stability properties of the different forms.

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Appendix: Steady-state solutions

In this Appendix, the steady-state solution for radial flows is derived based only on the assumption that the density, pressure, and velocity exist at some point. Therefore, it is representative of all smooth steady-state solutions in radial coordinates. Clain et al. [13] derived a differential equation for pressure under essentially the same assumptions which they then solved numerically. Here, the full solution is obtained analytically. Plots of their solution are qualitatively similar to the results given here.

The steady Euler equations with an axis of symmetry may be written as

$$\frac{\partial}{\partial r}\left(r^{\alpha}\rho u\right) = 0, \tag{80}$$

$$\frac{\partial}{\partial r}\left(r^{\alpha}\rho u^{2}\right) = -r^{\alpha}\frac{\partial P}{\partial r}, \tag{81}$$

$$\frac{\partial}{\partial r}\left(r^{\alpha}\rho H u\right) = 0, \tag{82}$$

where $\alpha = 0,\ 1,\ 2$ for Cartesian, cylindrical, or spherical coordinates, respectively. The variables are scaled by the values at a reference radius $r = r_{\mathrm{R}}$:

$$\begin{aligned} \tilde{r} &= r/r_{\mathrm{R}}, \qquad & \tilde{u} &= u(r)/u_{\mathrm{R}}, \\ \tilde{\rho} &= \rho(r)/\rho_{\mathrm{R}}, & \tilde{p} &= p(r)/p_{\mathrm{R}}, \end{aligned} \tag{83}$$

where a subscript of R indicates evaluation at $r = r_{\mathrm{R}}$. Assuming an ideal gas, the equations become (dropping the tildes immediately):

$$\frac{\partial}{\partial r}\left(r^{\alpha}\rho u\right) = 0, \tag{84}$$

$$\frac{\partial}{\partial r}\left(r^{\alpha}\rho u^{2}\right) + r^{\alpha}\mathrm{Eu}\frac{\partial P}{\partial r} = 0, \tag{85}$$

$$\frac{\partial}{\partial r}\left[r^{\alpha}\rho u\left(\frac{\gamma}{\gamma-1}\frac{p}{\rho}\mathrm{Eu} + \frac{u^{2}}{2}\right)\right] = 0, \tag{86}$$

subject to $u(1) = 1$, $p(1) = 1$, $\rho(1) = 1$, and where $\mathrm{Eu} = P_{\mathrm{R}}/(\rho_r u_{\mathrm{R}}^2)$. Integrating the first and third equation give:

$$r^{\alpha}\rho u = c_1, \tag{87}$$

$$\left(\frac{\gamma}{\gamma-1}\frac{P}{\rho}\mathrm{Eu} + \frac{u^{2}}{2}\right) = c_2. \tag{88}$$

Evaluating at $r = 1$:

$$c_1 = 1, \qquad c_2 = \left(\frac{\gamma}{\gamma-1}\mathrm{Eu} + \frac{1}{2}\right). \tag{89}$$

Solving for P in terms of c_2 and using $\rho = 1/(r^{\alpha}u)$:

$$P = \frac{\gamma-1}{\gamma\mathrm{Eu}}\frac{1}{r^{\alpha}u}\left(c_2 - \frac{u^2}{2}\right). \tag{90}$$

Substituting into the momentum equation (85), and using $r^{\alpha}\rho u = 1$ yields

$$\frac{\mathrm{d}u}{\mathrm{d}r} = -r^{\alpha}\mathrm{Eu}\frac{\mathrm{d}}{\mathrm{d}r}\left[\frac{\gamma-1}{\gamma\mathrm{Eu}}\frac{1}{r^{\alpha}u}\left(c_2 - \frac{u^2}{2}\right)\right]. \tag{91}$$

Separating variables, integrating and applying the boundary conditions

$$\frac{1}{u}\left(\frac{2c_2-1}{2c_2-u^2}\right)^{\frac{1}{\gamma-1}} = r^{\alpha}. \tag{92}$$

This gives u implicitly as a function of r. However, r can be found explicitly as a function of u. Hence, for a range of values of u, the corresponding values of r may be found and then the density and pressure are given by

$$\rho = \left(\frac{2c_2-u^2}{2c_2-1}\right)^{\frac{1}{\gamma-1}}, \tag{93}$$

$$P = \frac{\gamma-1}{2\gamma\mathrm{Eu}}\frac{\left(2c_2-u^2\right)^{\frac{\gamma+1}{\gamma-1}}}{(2c_2-1)^{\frac{1}{\gamma-1}}}. \tag{94}$$

In order to find the interval of existence of the solution, writing (92) as

$$u\left(2c_2-u^2\right)^{\frac{1}{\gamma-1}} = r^{-\alpha}\,(2c_2-1)^{\frac{1}{\gamma-1}}, \tag{95}$$

and differentiating implicitly led to

$$\begin{aligned} &\frac{\mathrm{d}u}{\mathrm{d}r}\left(2c_2-u^2\right)^{\frac{2-\gamma}{\gamma-1}}\left[2c_2 - u^2\left(\frac{\gamma+1}{\gamma-1}\right)\right] \\ &\quad = \alpha r^{-(\alpha+1)}(2c_2-1)^{\frac{1}{\gamma-1}}. \end{aligned} \tag{96}$$

The quantity in square brackets is more restrictive on the values of u than the factor before it. Using the definition of c_2, we find that (in the unscaled variables)

$$u^2 < 2\frac{\gamma-1}{\gamma+1}H_{\mathrm{R}} = a^{*}, \tag{97}$$

where H_{R} is the enthalpy evaluated at $r = R$ (or any other point, since H is constant) and a^{*} is the critical speed of

sound. In particular, $2c_2 - u^2 \neq 0$ is equivalent to requiring the internal energy be positive, while

$$2c_2 - u^2 \frac{\gamma + 1}{\gamma - 1} \neq 0, \tag{98}$$

is equivalent to requiring u to be subsonic.

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